Section XII

THE ABDOMEN & PELVIS

Obscurity is painful to the mind, as well as the eye.

David Hume

Chapter 39

Liver Failure

Intellectually, we stand on an islet in the midst of an illimitable ocean of inexplicability. Our business . . . is to reclaim a little more land.

T.H. Huxley (*a*)

The management of patients with liver failure is a particular challenge because of the numerous life-supporting functions that are lost when the liver fails. These include: (*a*) the production of about 20,000 proteins, including all the coagulation factors and transport proteins in blood, (*b*) the metabolism of innumerable endogenous and exogenous substances, and (*c*) the removal of microbes that breach the mucosal barrier in the GI tract.

This chapter describes the major problems that prompt ICU admission in patients with liver failure, and includes the problems associated with both acute liver failure (with no prior history of liver disease) and decompensated cirrhosis. This is an expansive topic that is only briefly described here, and a number of clinical practice guidelines are included in the bibliography at the end of the chapter (1–5).

ACUTE LIVER FAILURE

Acute liver failure (ALF) is an abrupt and rapid deterioration in liver function that occurs *de novo*, without prior liver disease. This is an uncommon condition, as demonstrated in Figure 39.1 by the number of cases (2,614) recorded in the United States over a period of 20 years (6).

Etiologies

The conditions included in Figure 39.1 were responsible for about 80% of cases of ALF in the United States, and acetaminophen tops the list by a great margin.

Acetaminophen

Acetaminophen is the leading cause of ALF in both the United States and the United Kingdom (being responsible for 46% and 64% of cases, respectively) (6). Acetaminophen hepatotoxicity is described in detail in Chapter 52, and is only briefly reviewed here. The source of the liver injury is depletion of hepatic glutathione stores, which results in oxidative cell injury from an

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acetaminophen metabolite. (Glutathione is the major intracellular antioxidant in the body, as described in Chapter 25.) The liver injury can begin as early as 12 hours after ingestion, and it peaks in 3 to 5 days (7). The laboratory abnormalities in the early stages of injury are characterized by high alanine aminotransferase (ALT) levels, with only mild increases in bilirubin (6). Acetaminophen-induced liver injury has two unfortunate features:

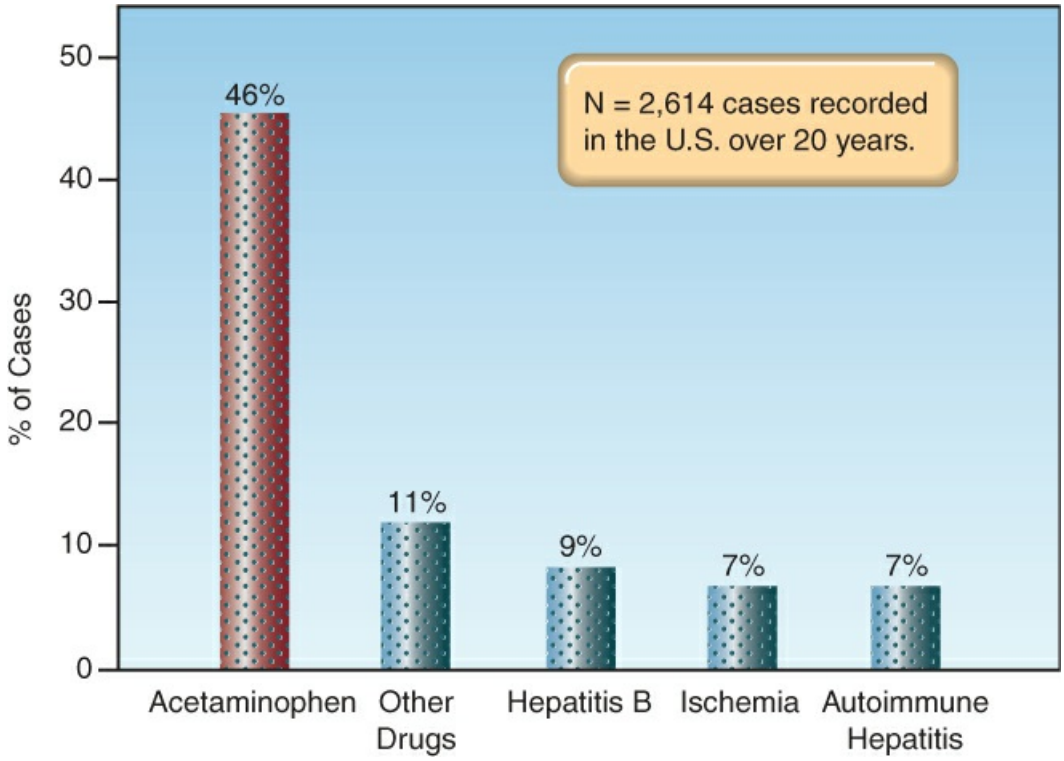


FIGURE 39.1 Causes of acute liver failure recorded by the United States Adult Acute Liver Failure Registry over the time period from Jan, 1998 to March, 2019. Adapted from Reference 6.

. About half of acetaminophen overdoses are unintentional (7), and are the result of attempts to alleviate pain. This is partly due to consumer ignorance about the potential toxicity of acetaminophen, and is partly due to the inclusion of acetaminophen in more than 600 over-the-counter products.

. Acetaminophen is undetectable in the plasma in about 50% of the cases where it causes liver injury (8). This creates a risk of missed diagnoses and poor outcomes, since the condition is treatable.

N-ACETYLCYSTEINE: The antidote for acetaminophen hepatotoxicity is N-acetylcysteine (NAC), which acts as a glutathione surrogate (see Figure 25.7). It is usually given intravenously (see Chapter 52 for the dosing regimen), and is started as soon as an acetaminophen overdose is confirmed or suspected. It is most effective when treatment is started within 10 hours of ingestion (7), and the efficacy declines with time thereafter. However, NAC is started at any time after a suspected or confirmed overdose of acetaminophen, and treatment is continued until there is evidence of improvement in liver function. About 75% of cases resolve without the need for liver transplantation (6).

The success of NAC in acetaminophen-induced ALF has led to its use in cases of ALF that are not linked to acetaminophen. Although there is no convincing evidence of benefit (9), it is considered reasonable to use NAC empirically in ALF of undetermined etiology (10).

Other Drugs

The drugs that have been reported to cause acute liver injury are listed in Table 39.1 (11). Drug-induced liver injury can be dose-related or idiosyncratic, and is more likely to occur in patients who are elderly, debilitated, or alcoholic. The idiosyncratic reactions are the most troublesome, with a survival rate of only 40% without liver transplantation (6).

TABLE 39.1 Drugs Associated with Acute Liver Failure†

Dose-Related	Idiosyncratic
Acetaminophen	Amiodarone§
Amiodarone§	Amoxicillin
Antimetabolites	Dantrolene
Antiretroviral Drugs	Halothane
Cyclosporine	Lisinopril
Statins§	Phenytoin
Valproic Acid	Statins§
	Sulfonamides

†For a more complete list, see References 6 and 11.

§Hepatic injury can be dose-related or idiosyncratic.

Ischemic Injury

Ischemic liver injury (sometimes called ischemic hepatitis), typically follows a period of systemic hypoperfusion (e.g., from hemorrhagic shock), and is characterized by very high transaminase levels (e.g., >3,000 IU/L) with only mild increases in bilirubin (<5 mg/dL) (6). This condition tends to resolve if the hypoperfusion resolves, but about 25% of cases need liver transplantation (6).

Acute Hepatitis

Autoimmune hepatitis has a more insidious onset than the other sources of ALF, and the diagnosis often rests on a clinical history of autoimmune disease, or positive serologies (e.g., antinuclear antibodies or anti-smooth muscle antibodies). About 80% of cases are in women (6), and hypergammaglobulinemia is common.

Viral hepatitis is responsible for <10% of cases of ALF in developed countries (5). Most cases are caused by the hepatitis B virus, which can be a new infection, or a latent infection that is sparked by immunosuppressive therapy. Once liver failure develops, treatment (with nucleoside analogues) is of little value (12), and survival is only 25% without liver transplantation (6).

Other Causes

Causes of ALF that are not included in Figure 39.1 include the HELLP syndrome (hemolysis, elevated liver enzymes and low platelets in pregnant and post-partum women), Budd Chiari

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syndrome (hepatic vein occlusion), other viral infections (e.g., herpes simplex), and infiltrating neoplasms (6).

Clinical Features

Unlike the liver failure associated with cirrhosis, ALF is not accompanied by ascites or bleeding from esophageal varices. Instead, the clinical picture is dominated by systemic inflammation (originating from the damaged liver) with hemodynamic instability and multiorgan dysfunction (similar to inflammatory shock, as described in Chapter 17).

Hemodynamics

The hemodynamic instability in ALF is due to vasodilation, and an ineffective circulating blood volume from hypoalbuminemia (10). The management involves volume resuscitation and vasopressors if needed, and *albumin is recommended as the resuscitation fluid*, especially when the serum albumin is below 3 g/dL (2). Albumin is available as a 5% solution that has an albumin concentration of 5 g/dL (similar to the normal concentration in plasma), and about 70% of the infused solution will remain in the intravascular space (see Figure 10.1). The use of *Ringer's lactate is not advised* because it promotes hyponatremia, which will exacerbate the cerebral edema in hepatic encephalopathy.

Acute Kidney Injury

Acute kidney injury (AKI) is reported in 70% of patients with ALF, and is usually seen in patients with acetaminophen hepatotoxicity, or those with hypotension who require vasopressors (13). Renal replacement therapy is used more aggressively in ALF because it protects against hepatic encephalopathy by removing ammonia as well as excess volume (see next).

Hepatic Encephalopathy

Encephalopathy is the dominant feature of acute liver failure, and is characterized by cerebral edema (as shown in Figure 39.2).

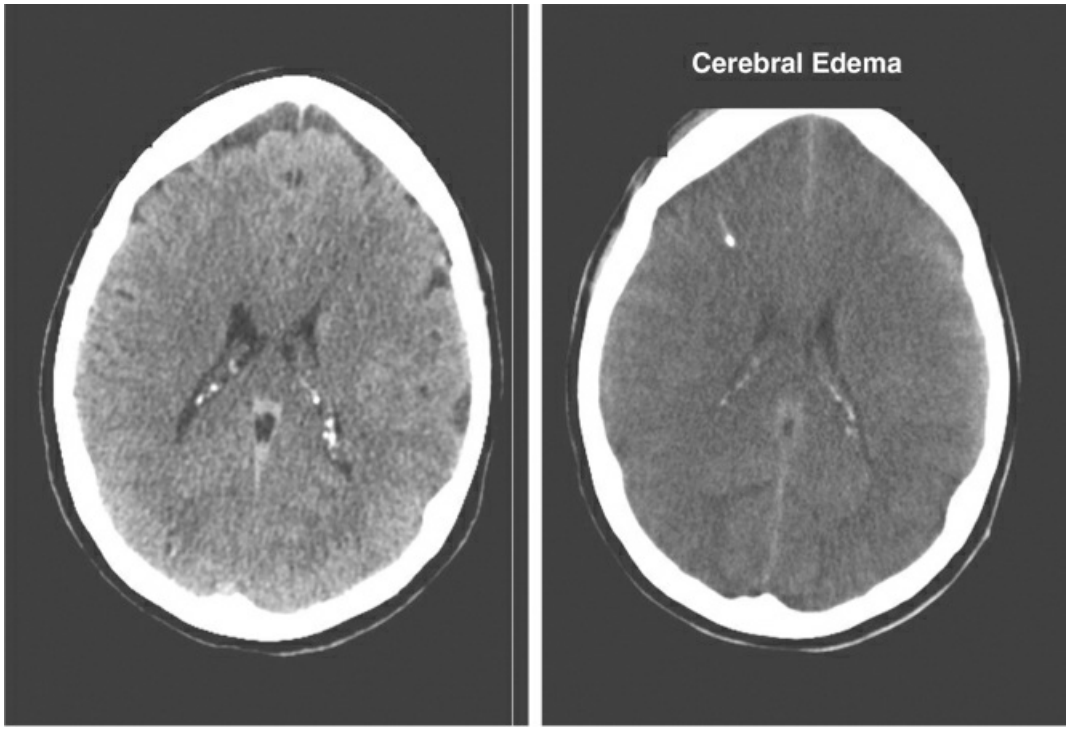


FIGURE 39.2 CT images of the head in a patient with acute liver failure, taken before (image on the left) and after (image on the right) the onset of cerebral edema. Note the effacement of sulci and loss of the grey-white interface in the image on the right. Images from Reference 6.

Pathogenesis

Ammonia has an important role in the cerebral edema that can develop in liver failure (14). Ammonia is a byproduct of protein degradation, and is produced primarily in the bowel (and to a lesser degree in skeletal muscle and kidneys). The liver plays a major role in clearing ammonia by converting it to urea via the urea cycle. This clearance mechanism is impaired or lost in liver failure, resulting in a progressive rise in ammonia levels in blood. Ammonia eventually crosses the blood-brain barrier and is taken up by astrocytes, where it is used to produce glutamine. The subsequent accumulation of glutamine then creates an osmotic force that draws water into the astrocytes and creates *a "cytotoxic" type of cerebral edema*.

Clinical Features

The clinical manifestations of hepatic encephalopathy include both cognitive and motor deficits. The cognitive changes that appear in progressive stages of hepatic encephalopathy are shown in Table 39.2 (15). Personality changes and mild disorientation are prominent in the early stages, while depressed consciousness dominates the later stages. The motor deficits can appear at any stage, and include extrapyramidal and cerebellar signs like rigidity, dysphagia, ataxia, and a flapping tremor (asterixis).

TABLE 39.2 The "West Haven" Criteria for Hepatic Encephalopathy

Grade	Features
Grade 1	• Short attention span

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	• Personality change
Grade 2	• Apathy or lethargy • Disoriented to time
Grade 3	• Stuporous but responsive • Disoriented to place and time
Grade 4	• Coma

From Reference 15.

PLASMA AMMONIA LEVEL: Plasma ammonia levels are almost always elevated in patients with hepatic encephalopathy, and in many (but not all) cases, there is a correlation between the degree of elevation of the plasma ammonia and the severity of the encephalopathy (16). (This correlation occurs more frequently in ALF than in decompensated cirrhosis.) Arterial ammonia levels may be more predictive than venous levels, at least in the early stages of encephalopathy (16). *An arterial ammonia level >150 µmol/L (normal <60 µmol/L) marks an increased risk of cerebral edema* (17).

Management

The prevention and treatment of hepatic encephalopathy is aimed at reducing the ammonia burden in the central nervous system. This is done by reducing ammonia production in the bowel (e.g., with lactulose and a nonabsorbable antibiotic), and clearing ammonia from the bloodstream (with hemofiltration).

LACTULOSE: Lactulose is a disaccharide that is metabolized by "lactic acid bacteria" (e.g., *Lactobacillus acidophilus*) in the bowel (18). This promotes the formation of short-chain fatty acids, and the resulting acidification of the bowel lumen helps to eradicate ammonia-producing microbes. (The bactericidal actions of an acid pH are illustrated in Figure 4.3.) The dosing recommendations for lactulose are as follows (19):

. *Oral or Nasogastric Route:* Start with 20–30 g (30–45 mL) every hour until laxation occurs, then reduce to 20 g (30 mL) every 8 hours, and titrate to achieve 2–3 loose stools daily.

. *Retention Enema:* Mix 200 g (300 mL) with 700 mL of tap water. Administer by high rectal enema, and retain for 30–60 minutes with patient in the Trendelenburg position.

Lactulose can produce an osmotic diarrhea, and the dosage should be reduced (or temporarily halted) if diarrhea appears.

RIFAXIMIN: Rifaximin is a nonabsorbable antibiotic with a wide spectrum of activity that has proven successful in alleviating symptoms of hepatic encephalopathy when given in a dose of 400 mg three times daily for at least 10 days (20). Although effective when used alone, it is usually combined with lactulose.

HEMOFILTRATION: Continuous venovenous hemofiltration (CVVH) can lower plasma ammonia levels and improve outcomes in patients with ALF (21), especially with continued use (20). At

the present time, CVVH is considered superior to lactulose or rifaximin for patients with ALF and hepatic encephalopathy, and should be considered in the following situations: (*a*) persistent oliguria, (*b*) plasma ammonia level >150 μmol/L, and (*c*) evidence of cerebral edema, regardless of the serum creatinine (5). CVVH should be started as soon as possible, as its effectiveness is time-dependent (22).

Liver Transplantation

Liver transplantation is the definitive treatment for ALF, and has reduced the mortality rate from 80% to 33% (6). There are several scoring systems that are used to identify who will benefit from transplantation, but this is the domain of the transplant specialist, who should be consulted for any patient with ALF who develops progressive hepatic encephalopathy.

ACUTE-ON-CHRONIC LIVER FAILURE

Most admissions to the ICU for liver failure involve patients with chronic liver disease (usually cirrhosis) who experience an abrupt deterioration in liver function. This *acute-on-chronic liver failure* is often complicated by: (*a*) bleeding from gastroesophageal varices, (*b*) hepatic encephalopathy, (*c*) infections, and (*d*) acute kidney injury. This section describes each of these complications, with the exception of hepatic encephalopathy, which is described in the previous section.

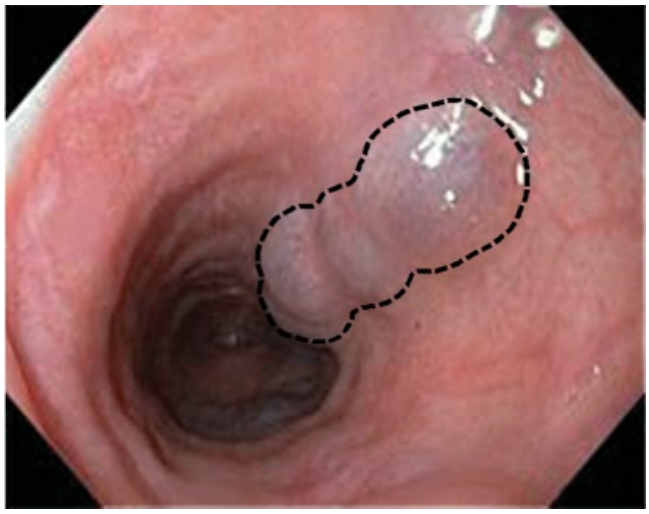


FIGURE 39.3 Endoscopic view of esophageal varices, which appear as engorged submucosal veins (outlined by the dotted line). Image courtesy of S. Bhimji, MD; digitally retouched.

Variceal Hemorrhage

Gastroesophageal varices are engorged submucosal veins that appear as a consequence of portal hypertension (see Figure 39.3). Acute bleeding from ruptured varices is one of the more life-threatening complications of advanced liver disease.

Management

The management of acute variceal bleeding, which is outlined in Table 39.3 (4), involves source

control (with endoscopy), resuscitation (with fluids and blood products), vasoactive therapy (with octreotide), and prophylactic measures (with antibiotics). Intubation may also be necessary in patients with altered mentation.

TABLE 39.3 The Management of Acute Variceal Bleeding

Component	Recommendations
Endoscopy	• Perform within 12 hrs.
Volume Resuscitation	• Maintain MAP of 65 mm Hg. • Use 5% albumin as a resuscitation fluid if serum albumin <3 g/dL. • Avoid hypotonic fluids.
Blood Products	• Transfuse RBCs, plasma, and platelets in a 1:1:1 ratio. • Use a target Hb of 7 g/dL. • Use thromboelastography to guide hemostatic therapy (see Table 15.1).
Splanchnic Vasoconstrictor	• Octreotide: 50 µg as IV bolus, then infuse at 50 µg/hr for 2–5 days.
Antibiotic Prophylaxis	• Ceftriaxone, 1 g IV daily for 5–7 days. • First dose before endoscopy.

From the guidelines in References 1 and 4.

Endoscopy is recommended within 12 hours of presentation (1,4), both to identify the source of the bleeding (which has prognostic value) and to clip or band the bleeding varix. The resuscitation with asanguinous fluids and blood products is the same as described for hemorrhagic shock in Chapter 15, with the following exceptions:

. Albumin is recommended as a resuscitation fluid, especially when the serum albumin is less than 3 g/dL (2). The 5% albumin solution has an albumin concentration of 5 g/dL, which is close to the normal albumin concentration in plasma, and about 70% of the infused volume will expand the plasma volume.

. Hypotonic fluids like Ringer's lactate are not recommended because they add to the risk of cerebral edema (which is responsible for hepatic encephalopathy).

. The INR is not a valid measure of bleeding risk in patients with cirrhosis (4), and thromboelastography (described in Chapter 15) is recommended to guide hemostatic therapy (1).

The management of variceal bleeding also includes a splanchnic vasoconstrictor like *octreotide*, a somatostatin analogue that is started as soon as possible, using the dosing regimen in Table 39.3. Prophylactic antibiotics should also be started as soon as possible, and *ceftriaxone* (2 grams daily) is recommended in most guidelines (2,4). (*Note:* This recommendation assumes that resistant microbes are not prevalent in your ICU.)

Refractory Bleeding

Variceal bleeding is refractory to endoscopy and optimal medical management in about 20% of

cases (4). In this situation, a transesophageal portosystemic shunt (TIPS) procedure should be considered. TIPS is a noninvasive method that creates a shunt between the portal and hepatic veins, thereby alleviating the portal venous hypertension that creates varices. Unfortunately, the TIPS procedure trades one problem for another, as bypassing the liver increases the risk of hepatic encephalopathy.

Spontaneous Bacterial Peritonitis

About one-third of patients with decompensated cirrhosis have a bacterial infection, and the most common infection is *spontaneous bacterial peritonitis* (SBP), which is an infection of ascitic fluid without a primary site of infection in the abdomen (5). The source of this infection is the "translocation" of enteric pathogens across the bowel mucosa. SBP occurs only in patients with cirrhosis and ascites, because cirrhosis impairs the normal function of the liver to eradicate microbes that translocate across the bowel wall.

About 50% of cases of SBP are community-acquired, while 25% are classified as "healthcare-associated" (i.e., the patient has had contact with a healthcare facility within 90 days), and the remaining 25% are nosocomial (23). A single organism is isolated in most cases of SBP, and the most common isolates are gram-negative organisms (especially *Escherichia coli* and *Klebsiella pneumoniae*) with gram-positive organisms (usually staphylococci and enterococci) a distant second (23). Resistant organisms are isolated in 35% of cases worldwide, but the prevalence is much lower (16%) in the United States (23).

Diagnosis

SBP can present with fever, abdominal pain, and rebound tenderness, but it is asymptomatic in one-third of cases (24). A paracentesis is required for the diagnosis, and *an absolute neutrophil count >250/mm^3 in ascitic fluid secures the diagnosis, regardless of the culture result* (5). An organism is isolated in about 60% of cases (23). A positive culture without the required neutrophil count in ascitic fluid is not sufficient for the diagnosis of SBP (5). (In this situation, a repeat paracentesis is wise.)

SBP should be suspected in all patients with decompensated cirrhosis and liver failure who are admitted to the ICU, or who experience a sudden deterioration while in the hospital (5). A paracentesis should be performed when SBP is suspected, and the fluid sent for cell count and culture. For optimal culture results, the ascitic fluid should be added to blood culture bottles (aerobic and anaerobic) at the bedside.

Antibiotic Therapy

Antibiotic therapy is started in all cases when the absolute neutrophil count is >250/mm^3 (before the culture results are available). The choice of antibiotic is determined by the place of origin of the infection, and the prevalence of resistant organisms in your ICU. This is shown in the algorithm in Figure 39.4 (25), which can be summarized as follows:

. Resistant organisms are unlikely in community-acquired SBP (23), and a third generation cephalosporin (e.g., cefotaxime or ceftriaxone) is generally recommended.

. Resistant organisms are typically isolated in nosocomial cases of SBP, and less frequently in healthcare-associated infections (23). (*Note:* A nosocomial infection is one that appears after

48 hrs in the hospital, while a healthcare-associated infection is one that appears within 48 hrs after admission in a patient who has had contact with a healthcare facility in the last 90 days.)

a. If a resistant infection is possible, a carbapenem (e.g., meropenem 1 gram every 8 hours) is recommended (25), with vancomycin added if methicillin-resistant *Staphylococcus aureus* is a potential offender, or daptomycin if vancomycin-resistant enterococci are suspected.
b. If a resistant organism is unlikely, broad-spectrum coverage for enteric pathogens (e.g., with piperacillin-tazobactam) is sufficient.

Antibiotics should be continued for at least 7 days, regardless of the culture results. The response to antibiotics can be assessed by a repeat paracentesis after 48 hours: i.e., a favorable response is defined as a ≥25% decrease in the neutrophil count (5).

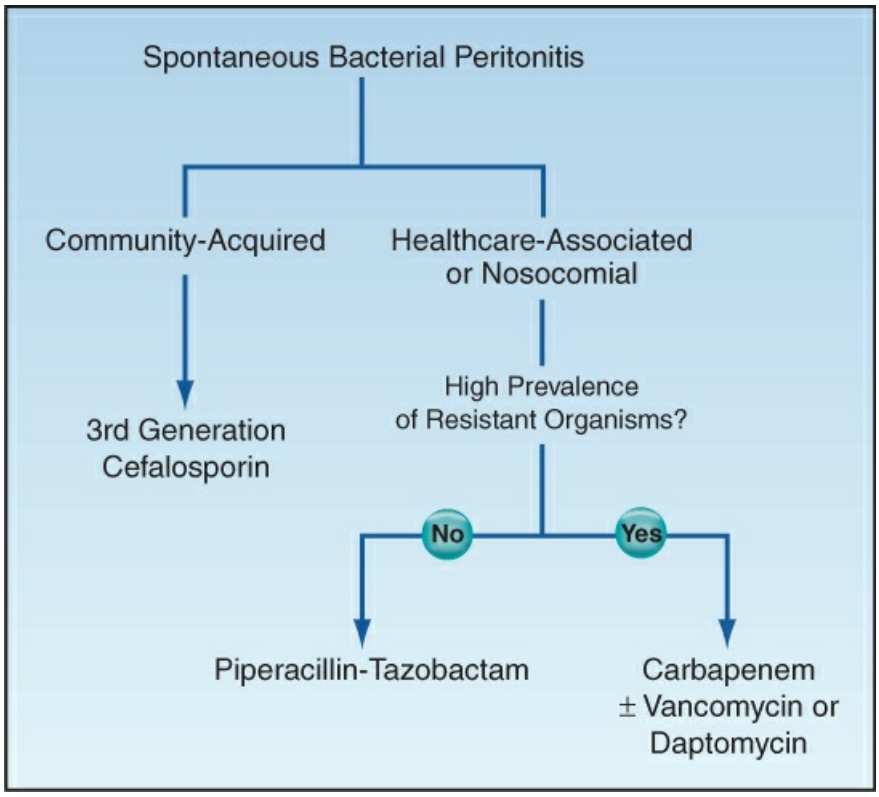


FIGURE 39.4 Algorithm for the empiric antibiotic treatment of spontaneous bacterial peritonitis. From Reference 25.

Albumin Infusions

Infusions of albumin have been shown to improve survival in SBP (26,27). This was originally attributed to an improvement in renal function (27), because acute kidney injury develops in 30% of patients with SBP (25), and this increases the mortality rate considerably (28). However, more recent studies have shown that the survival benefit of albumin is independent of the serum creatinine (26). The recommended albumin regimen is 1.5 g/kg on day one, followed by 1 g/kg on day 3 (5). The 25% albumin solution (25 g/100 mL) is recommended, to limit the infusion volume.

Hepatorenal Syndrome

Acute kidney injury (AKI) is reported in as many as 50% of patients with decompensated cirrhosis (5). The hepatorenal syndrome (HRS) is a type of "functional AKI" that occurs without intrinsic renal disease, and its appearance heralds a poor prognosis.

Pathogenesis

HRS is the result of hemodynamic alterations in the splanchnic and renal circulations. Cirrhosis is associated with splanchnic vasodilation, and the neurohumoral (renin system) response to this vasodilation results in vasoconstriction in other organs, including the kidneys. The renal vasoconstriction creates a situation where the glomerular filtration rate is vulnerable to small decrements in cardiac output. Sepsis enhances the vasodilation from cirrhosis, and HRS is more frequent when cirrhosis is accompanied by sepsis.

Diagnosis

The diagnostic criteria for HRS are listed in Table 39.4 (29). The criteria include evidence of AKI (i.e., an increase in serum creatinine ≥0.3 mg/dL within 48 hrs) with no other apparent cause (i.e., no circulatory shock or nephrotoxic drugs), and no evidence of intrinsic kidney disease. Urinary indices in HRS are similar to "prerenal" indices (i.e., low sodium and high osmolarity), and these could differentiate HRS from acute tubular necrosis.

TABLE 39.4 Diagnosis and Management of Hepatorenal Syndrome

I. *Diagnostic Criteria:*
1. Cirrhosis with ascites.
2. Increase in SCr of ≥0.3 mg/dL within 48 hrs, or SCr ≥1.5 times baseline.
3. No response to diuretic withdrawal and 2 days of fluid challenges with albumin (1 g/kg/day).
4. No evidence of shock, and no recent use of nephrotoxic drugs.
5. No signs of structural kidney injury:
   a. Absence of proteinuria (>500 mg/day).
   b. Absence of hematuria (>50 RBC's per high power field).
   c. Normal findings on renal ultrasound.

II. *Management:*
1. Albumin (25%): 20–40 g daily.
2. Splanchnic vasoconstriction with one of the following:
   a. Terlipressin: 1 mg IV every 4–6 hrs, or infuse at 84 µg/hr (2 mg/day).
   b. Norepinephrine: 5–40 µg/min by continuous infusion.
   c. Octreotide (100–200 µg subQ) plus midodrine (7.5–15 mg PO) three times daily.

From Reference 29.

Management

The management of HRS is included in Table 39.4. Daily doses of albumin are recommended at a dose of 20–40 grams daily, which should be given as a 25% albumin solution (25 grams per 100 mL) to limit the volume infused. The benefits of albumin may not be related to volume, as albumin is also an antioxidant, and has immunomodulating effects (30).

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The mainstay of treatment for HRS is to promote splanchnic vasoconstriction and reverse the hemodynamic changes that precipitate the condition. The drug of choice for this is *terlipressin*, a synthetic vasopressin analogue that acts primarily as a splanchnic vasoconstrictor. Clinical studies have shown that terlipressin improves renal function in HRS (31,32), but it does not improve survival or reduce the need for liver transplantation (32). Side effects of terlipressin include abdominal pain, diarrhea, pulmonary edema (hydrostatic), and ischemic events. The drug was approved for use in the United States in 2022.

Alternatives to terlipressin include *norepinephrine*, and the combination of *midodrine and octreotide* (33). (See Table 39.4 for dosing recommendations.) Norepinephrine has been equivalent to terlipressin in clinical trials (34), but the experience is limited, while midodrine plus octreotide is inferior to terlipressin (35), and is limited to use in countries where terlipressin is not available.

Increased Abdominal Pressure

Increased intra-abdominal pressure is an overlooked cause of AKI in critically ill patients (36), as described in Chapter 34, and it is possible that it could also promote HRS. This is supported by a study showing that large-volume paracentesis (i.e., removal of ≥4 liters of fluid) improved renal function in patients with tense ascites and HRS (37). Therefore, measurement of intra-abdominal pressure is advised for all patients with tense ascites and HRS (see Chapter 34 for the method of measuring intra-abdominal pressure), and large-volume paracentesis is encouraged if the abdominal pressure exceeds 12 cm H_2O (29). However, careful hemodynamic monitoring is needed after large-volume paracentesis, to avoid hemodynamic compromise.

Liver Transplantation

Liver transplantation is the definitive treatment for HRS, and a consultation for liver transplantation should be obtained whenever HRS is diagnosed.

A FINAL WORD

The Liver & Bowel Sepsis

One of the recurring themes in this book is the importance of the bowel as a source of systemic infection and inflammation in critically ill patients (see Chapters 4 and 41). Much of the attention in this area is directed at the breakdown of the mucosal barrier in the bowel, but there is a second line of defense in the event of mucosal breakdown. This is illustrated by the occurrence of spontaneous bacterial peritonitis in patients with liver failure; an infection that is caused by enteric pathogens that escape across the bowel mucosa. This infection demonstrates the importance of the liver as a second line of defense against microbes that breach the mucosal barrier in the bowel. In fact, about three-quarters of the reticuloendothelial system is located in the abdomen (mostly represented by the liver), where it is positioned to protect against the systemic spread of enteric pathogens.

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Chapter 40

Acute Pancreatitis

Medicine cures the man who is fated not to die.

Chinese Proverb

Acute pancreatitis is cited as one of the leading abdominal disorders that requires hospitalization (1), but most cases are not severe enough to warrant ICU admission. However, the cases that do require ICU admission are often (or quickly become) the sickest patients in the ICU.

This chapter describes the clinical features and management of acute pancreatitis, with emphasis on the severe cases that require ICU-level care. Management of severe pancreatits is challenging, because there is no specific treatment (other than removing any predisposing condition), and the overall mortality rate (about 25%) has not changed over the past two decades (2).

PATHOGENESIS & PRESENTATION

The pancreas houses two types of cells: (*a*) "islet" cells that have an endocrine function, and secrete insulin, glucagon, and somatostatin, and (*b*) "acinar" cells that have an exocrine function, and secrete digestive enzymes such as amylase, lipase, and trypsin. Acute pancreatitis is an inflammatory process that primarily affects the acinar (exocrine) cells. Disruption of these cells releases digestive enzymes, and proteolytic enzymes like trypsin can aggravate the pancreatic injury and promote "autodigestion" of the pancreas. The pancreatic injury triggers a systemic inflammatory response, which can lead to inflammatory injury in other organs, culminating in multiorgan failure.

Etiologies

The causes of acute pancreatitis are listed in Table 40.1. About three-quarters of the cases are the result of gallstones, alcohol abuse, or are idiopathic (1,3), while the other causes that deserve mention are hypertriglyceridemia (4,5), drugs (6), and infections (7).

Hypertriglyceridemia

Severe hypertriglyceridemia (serum triglyceride levels >1,000 mg/dL) is responsible for up to

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7% of cases of acute pancreatitis (4), and the frequency of cases has increased considerably in recent years (5). Several mechanisms may be involved, but one of the major ones is the release of polyunsaturated fatty acids (PUFA) from triglycerides, since PUFAs are highly oxidizable, and are capable of inciting an inflammatory response. (*Note:* The oxidation of PUFAs in food products is responsible for "rancidity".)

The pancreatitis from hypertriglyceridemia is more severe than other types of pancreatitis, and is associated with a greater degree of systemic inflammation and extrapancreatic organ injury (4). Fortunately, there is an effective treatment, which is described later in the chapter.

TABLE 40.1 Etiologies of Acute Pancreatitis

More Common†	Less Common
Gallstones (21–33%)	Abdominal Trauma
Alcohol (16–27%)	Hypercalcemia
Idiopathic (10–15%)	Infections
Hypertriglyceridemia (2–7%)	Pregnancy
Drugs (3–4%)	Vasculitis

†Reported prevalence indicated in parentheses.

Drugs

At least 36 drugs have been implicated as a causative factor in acute pancreatitis (1,6), but many of the cases are idiosyncratic reactions in case reports. The most frequently reported drugs are hydrochlorothiazide, azathioprine, and doxycycline (6), while the other implicated drugs include (in alphabetical order) acetaminophen, enalapril, furosemide, metronidazole, procainamide, trimethoprim-sulfamethoxazole, and valproic acid (1,6).

Infections

A variety of infectious agents can cause pancreatitis (7), including viruses (HIV, cytomegalovirus, varicella-zoster, herpes simplex, and hepatitis B), bacteria (Mycoplasma, Legionella, Leptospira, and Salmonella), fungi (Aspergillus) and parasites (Toxoplasma, Cryptosporidium). Suspicion of an infectious agent is based on the presence of other manifestations of the infection.

General Features

The general features of acute pancreatitis are summarized in Table 40.2. This is the revised Atlanta classification for acute pancreatitis (8), and it has three components: (*a*) a classification based on morphology, (*b*) diagnostic criteria, and (*c*) a severity of illness classification. Each of these is described next.

Morphology

There are two types of acute pancreatitis, based on the presence or absence of pancreatic necrosis.

. *Edematous pancreatitis* is characterized by diffuse inflammation of the pancreas, with no

evidence of pancreatic necrosis. This is the most common type of pancreatitis, and the clinical presentation is typically mild and self-limited, with no involvement of other organs (1,2).

. *Necrotizing pancreatitis* is characterized by areas of necrotic destruction in the pancreas, which can extend into the peripancreatic tissues. This condition can be complicated by local infection and extrapancreatric organ injury. The incidence of this type of pancreatitis varies in different reports, from as low as 5% (8) to as high as 40% (4).

TABLE 40.2 Classification of Acute Pancreatitis†

Feature	Definitions or Criteria
Morphology	1. Edematous Pancreatitis 2. Necrotizing Pancreatitis
Diagnostic Criteria	Any 2 of the following: 1. Characteristic abdominal pain 2. Serum amylase or lipase >3 times upper limit of normal 3. Characteristic imaging
Severity of Illness:	
A. Mild	A. No organ failure, and no local or systemic complications
B. Moderately Severe	B. Transient organ failure (<48 hrs), or local or systemic complications without persistent organ failure
C. Severe	C. Persistent organ failure (>48 hrs), single or multiple

†The revised Atlanta classification, from Reference 8.

Diagnostic Criteria

The diagnosis of acute pancreatitis requires two of the following: (*a*) abdominal pain that is characteristic of acute pancreatitis, (*b*) an increase in serum amylase or lipase levels to three times the normal range, and (*c*) a contrast enhanced CT scan (or MRI) showing edema or necrotic lesions in the pancreas.

Pancreatic Enzymes

Although no distinction is made between amylase and lipase in the diagnostic criteria, *lipase should be favored over amylase* for the following reasons:

. An increase in serum lipase to three times normal has a greater specificity for acute pancreatitis than a similar increase in serum amylase (9).

. The elevation in lipase occurs earlier than the elevation in amylase (4–8 hrs vs. 6–12 hrs, respectively), and it lasts much longer (i.e., 8–14 days for lipase, versus 3–5 days for amylase) (9).

. Monitoring the lipase level alone has the same diagnostic accuracy as monitoring both (amylase and lipase) levels (9).

The combination of abdominal pain and increased pancreatic enzymes is diagnostic in about 75–80% of cases of acute pancreatitis (1), but imaging studies are a valued addition (see next).

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Imaging

CT scans of the abdomen (contrast-enhanced) are often included in the diagnostic evaluation of pancreatitis because they can identify the type of pancreatitis (edematous vs. necrotizing) and reveal localized complications (e.g., infection). Figure 40.1 shows a contrast-enhanced CT image of edematous pancreatitis. The pancreas is thickened and enhances completely, and the border of the pancreas is blurred, which is characteristic of pancreatic edema. Compare this with the image in Figure 40.2, which shows a large area in the pancreas that is not contrast-enhanced. This represents pancreatic necrosis, and identifies the condition as necrotizing pancreatitis. Pancreatic necrosis may not appear for the first week after the onset of symptoms (1), so repeat imaging is advised in patients with persistent symptoms or severe pancreatitis.

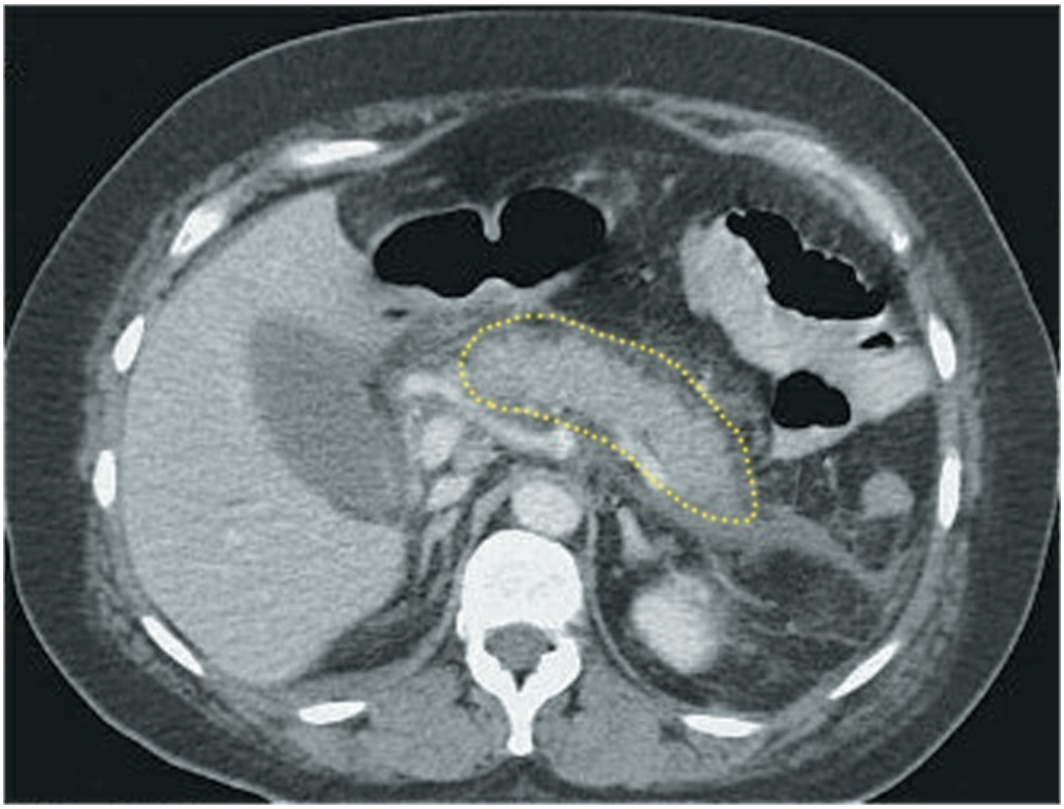


FIGURE 40.1 Contrast-enhanced CT image showing edematous pancreatitis. The pancreas (outlined by the dotted line) is enlarged and enhances completely. There is also blurring of the pancreatic border, which is characteristic of edema formation.

When IV contrast cannot be administered (e.g., because of a dye allergy), CT imaging is less likely to distinguish between edematous and necrotizing pancreatitis. In this situation, magnetic resonance imaging (MRI) is a suitable alternative.

Severity of Disease

The revised Atlanta classification of acute pancreatitis includes three categories for severity of disease, based on the presence or absence of complications, and extrapancreatic organ failure (see Table 40.2). About 50% of cases are classified as *mild disease* (symptoms self-limited, and no evidence of complications or organ failure), while the remaining 50% are equally divided

between *moderately severe disease* (transient organ failure, resolves in less than 48 hours) and *severe disease* (organ failure persists for longer than 48 hours) (10). The mortality in acute pancreatitis is mostly limited to patients with severe disease, and is about 20–25% (1,2). *Pancreatic enzymes and CT images have a poor correlation with the severity of illness.*

The extrapancreatic organ involvement in acute pancreatitis is related to the systemic inflammatory response, and can involve single or multiple organs. The organs typically involved are the lungs (i.e., acute respiratory distress syndrome), the kidneys (acute kidney injury), and the circulatory system (hypotension and circulatory shock). This organ involvement may not appear in the early stages of illness.

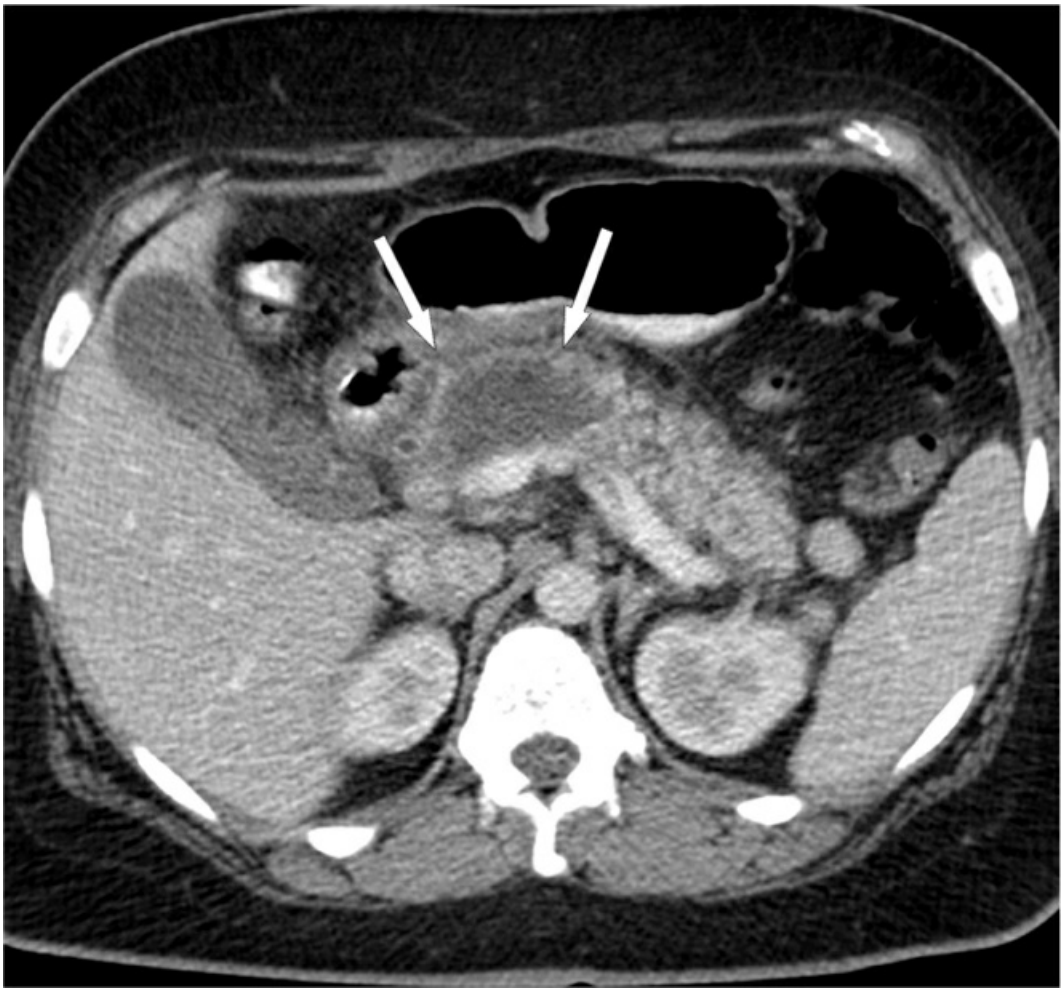


FIGURE 40.2 Contrast-enhanced CT image showing necrotizing pancreatitis. The area that is not contrast-enhanced (indicated by the arrows) represents necrosis in the neck and body of the pancreas. Image from Reference 8.

Biliary Evaluation

Since gallstones are the leading cause of acute pancreatitis, an evaluation of the gall bladder and biliary tree is advised in all cases of confirmed acute pancreatitis. Contrast-enhanced CT images may suffice for this evaluation, but in cases where a CT scan is unavailable or inconclusive, bedside ultrasonography is recommended.

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MANAGEMENT

Management in the ICU is primarily for patients with moderately severe or severe disease, and mostly involves supportive care, especially with fluids and nutrition.

Fluid Management

Severe pancreatitis is associated with loss of intravascular fluid through leaky systemic capillaries, and the resulting hypovolemia can produce additional pancreatic necrosis. For this reason, attention to volume resuscitation is recommended early in the course of severe pancreatitis. The following fluid regimen is recommended:

. Ringer's lactate (RL) is the recommended fluid, based on evidence of improved outcomes with RL compared to isotonic saline (11,12). However, since RL contains calcium, it should not be used in hypercalcemia-related pancreatitis.

. Start with a bolus infusion of 10 mL/kg in patients with hypovolemia, and follow with a continuous infusion of 1.5 mL/kg per hour (13). Avoid the bolus infusion if there is no evidence of hypovolemia.

. *Caution:* More aggressive volume infusion (i.e., bolus of 20 mL/kg, and a rate of 3 mL/kg/hr) promotes fluid overload, and does not improve outcomes (13).

. The goal of fluid management is to maintain a mean arterial pressure ≥65 mm Hg, and a urine output ≥0.5 mL/kg per hour.

Vasopressors

As always, vasopressors are advised if fluid therapy does not achieve the desired blood pressure. There are no official recommendations regarding vasopressor therapy in severe pancreatitis, but norepinephrine is an appropriate choice. All vasoconstrictor drugs can reduce splanchnic blood flow (especially phenylephrine), which could promote pancreatic necrosis, so careful titration of infusion rates (and avoidance of phenylephrine) is advised.

Nutrition

Nutrition should be started early (within 48 hours after the onset of illness) using an oral diet (if tolerated) or enteral tube feedings (1,14). Parenteral nutrition is reserved for cases where enteral feeding is not tolerated (e.g., from an ileus).

Enteral Nutrition

The preference for enteral nutrition is based on the ability of oral feeding to exert a trophic effect on the bowel mucosa. This helps to maintain the integrity of the bowel mucosa and reduce the risk of bacterial translocation across the bowel wall, which is considered the major source of pancreatic infections (15). Clinical studies have shown that *enteral nutrition is associated with fewer infections, less multiorgan failure, and a lower mortality rate than total parenteral nutrition* in patients with severe pancreatitis (16).

When enteral tube feedings are used, it is unclear if gastric or jejunal feeding is optimal (17). Based on evidence that nasogastric feeding produces no apparent harm in severe pancreatitis (18), it is reasonable to start with nasogastric feeding. Elemental feeding formulas (which are

low in fat) should be preferred.

Abdominal Complications

Local complications of pancreatitis include infection, pseudocyst formation, and abdominal compartment syndrome.

Pancreatic Infection

About one-third of patients with necrotizing pancreatitis develop an infection in the necrotic areas of the pancreas, which typically appears 7–10 days after the onset of illness (19). Infection may be heralded by persistent or recurrent fever and leukocytosis, but these findings are also common in necrotic pancreatitis without infection. A contrast-enhanced CT image can reveal probable infection by the presence of gas bubbles, as shown in Figure 40.3.

If infection is suspected, antibiotic therapy should be started, with coverage for gram-negative enteric pathogens. (*Note:* Prophylactic antibiotics do not reduce the incidence of pancreatic infections, and are not recommended (20).) However, these infections are difficult to treat with antibiotics, and CT-guided drainage may be necessary. There is some debate about the timing of drainage, but there is evidence that early drainage produces better results than delayed drainage (21). If noninvasive drainage is not successful, then surgical debridement (necrosectomy) may be necessary.

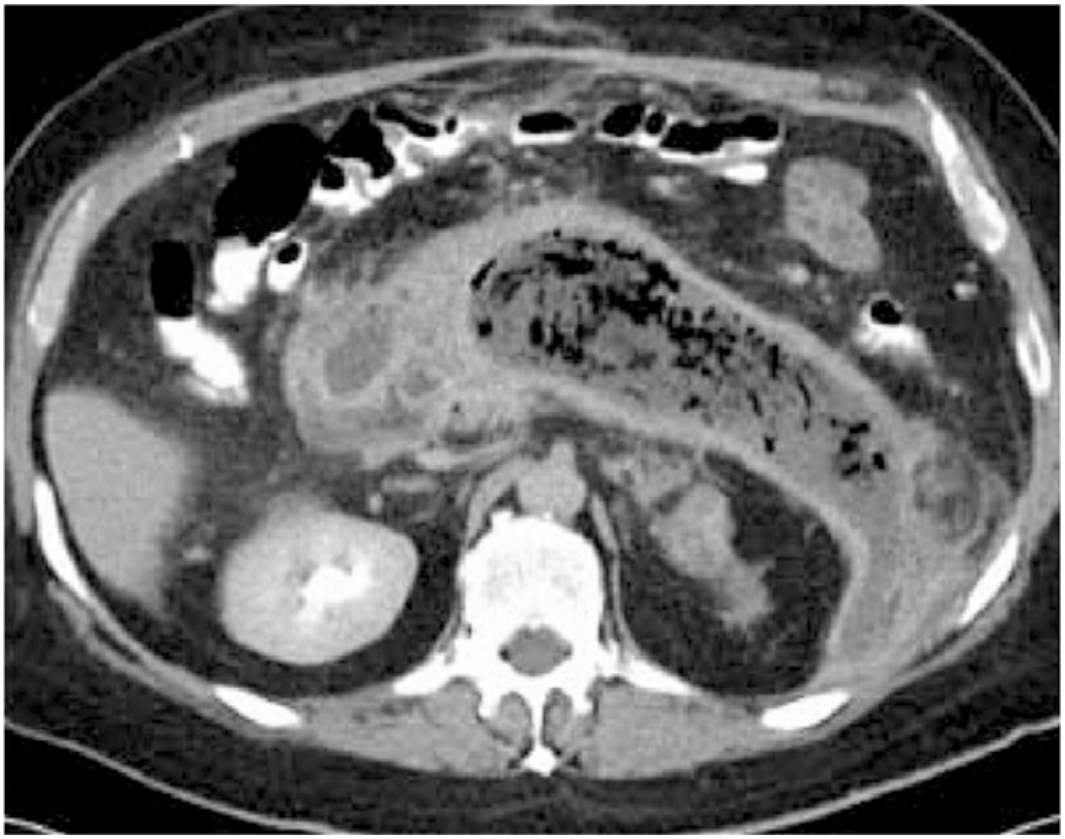


FIGURE 40.3 Contrast-enhanced CT image showing extensive necrosis of the pancreas with numerous gas bubbles, indicating infection.

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Abdominal Compartment Syndrome

There are several sources of increased intra-abdominal pressure in severe pancreatitis, including peripancreatic fluid collections, ascites, and edema of the bowel wall (from aggressive volume infusion). An increase in intra-abdominal pressure (>12 cm H_2O) has been reported in 55% of patients with severe pancreatitis (22), and pressures that exceed 20 cm H_2O can cause oliguria (as described in Chapter 34). Therefore, an abdominal pressure measurement is recommended for any patient with severe pancreatitis complicated by oliguria.

Specific Therapies

The following management is specific to the cause of the pancreatitis.

Hypertriglyceridemia

The pancreatitis associated with hypertriglyceridemia can be treated with insulin, which acts on lipoprotein lipase to reduce serum triglyceride levels (4). An intravenous infusion of insulin (0.1–0.4 Units/kg/hr) can reduce triglyceride levels by 85–90% in 24 hours if combined with fasting (23). The goal of treatment in acute pancreatitis is a serum triglyceride below 500 mg/dL (4). A 5–10% dextrose infusion is advised during the insulin infusion, along with careful monitoring of serum glucose levels. Once the pancreatitis has been successfully managed, continued treatment with lipid-lowering drugs like gemfibrozil or fenofibric acid may be needed to keep triglyceride levels below 500 mg/dL (4).

Gallstone Pancreatitis

When acute pancreatitis is associated with gallstones, endoscopic retrograde cholangio-pancreatography (ERCP) with sphincterotomy is indicated if there is evidence of biliary obstruction (e.g., by ultrasound). Cholecystectomy can be performed during the initial hospitalization for mild pancreatitis, but it is delayed (4–6 weeks) after an episode of severe pancreatitis.

A FINAL WORD

Alcohol & Pancreatitis (?)

Although alcohol is widely regarded as a major source of pancreatitis, the strong link between alcohol and pancreatitis was made at a time when the diagnosis of pancreatitis was based solely on elevation of the serum amylase. The problem is that amylase is also secreted by the parotid glands, and alcohol stimulates the release of salivary amylase. In fact, *hyperamylasemia of salivary origin is reported in 40% of cases of acute alcohol intoxication* (6). It is thus possible that elevations of salivary amylase created the strong association of alcohol abuse and pancreatitis.

There are about 30 million alcoholics in the United States (24), and the number of hospital admissions for pancreatitis is 275,000 per year (25), or about 69,000 per year if you assume that 25% of cases of acute pancreatitis are caused by alcohol. This means that less than 1% (0.2%) of alcoholics develop acute pancreatitis each year, which is not a very strong association.

Why is this important? Because, whenever a patient with acute pancreatitis has a history of alcohol abuse, no efforts are made to find another cause of the illness. It seems wise not to continue this behavior.

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Chapter 41

Abdominal Infections in the ICU

My dear Watson, you see, but you do not observe.

Arthur Conan Doyle (*a*)

The concept of the bowel as noxious reservoir first appeared in the early years of the twentieth century, when a Scottish surgeon named William Arbuthnot-Lane began performing total colectomies in patients with chronic constipation, to prevent "autointoxication" from toxic bowel contents (1). This practice was eventually abandoned (along with the surgeon), but the concept of autointoxication has been revived, as the bowel is now recognized as a leading source of morbidity and mortality in critically ill patients.

This chapter describes abdominal infections that are related to the ICU stay, including infections of the gallbladder (acalculous cholecystitis), the bowel (*Clostridium diffcile* enterocolitis), and the peritoneal cavity (e.g., postoperative infections).

ACALCULOUS CHOLECYSTITIS

Acalculous cholecystitis (i.e., inflammation of the gallbladder without cystic duct obstruction by gallstones) is an uncommon form of acute cholecystitis that typically occurs in critically ill patients, and is more severe than its "calculous" counterpart (2,3).

Pathogenesis

Predisposing conditions include stroke, sepsis, circulatory shock, prolonged periods of bowel rest (i.e., during total parenteral nutrition), and the postoperative period (especially following cardiopulmonary bypass surgery) (3–6). Possible mechanisms include hypoperfusion, decreased gallbladder contractions (which allows the buildup of gallbladder "sludge"), relaxation of the sphincter of Oddi (which allows enteric pathogens to enter the biliary tree), and a change in the composition of bile. The latter mechanism is based on the observation that biliary sludge contains small crystals called "microliths" that can promote cholecystitis (7).

Clinical Features

The clinical presentation can be nonspecific, as fever, hypotension, and multiorgan failure are

common but nonspecific findings, while *pain and tenderness in the right upper quadrant can be absent in one-third of patients* (8). Blood cultures are positive in 90% of cases (8) and gram-negative aerobic bacilli are isolated in almost all cases.

Diagnosis

Ultrasound is favored as the initial diagnostic test for acalculous cholecystitis, but the diagnostic yield can be limited. Gallbladder distension and sludge are suggestive findings, but are nonspecific. Advanced disease may yield more specific findings, such as the ultrasound image in Figure 41.1, which shows sloughed mucosa in the lumen of the gallbladder that is characteristic of gangrenous cholecystitis.

The diagnostic yield from ultrasound varies widely in different reports; the pooled results from 26 studies show a sensitivity of 81% and a specificity of 83% (9). If ultrasound is not diagnostic, *the gold standard test for acalculous cholecystitis is the hepatobiliary iminodiacetic acid (HIDA) scan* (2), but this requires a functional liver to move the tracer into the bile ducts. Magnetic resonance imaging (MRI) has the same diagnostic yield as abdominal ultrasound, and the accuracy of computed tomography is poor (2).



FIGURE 41.1 Transverse sonogram of the gallbladder showing marked thickening of the gallbladder wall and an echogenic mass projecting into the lumen of the gallbladder. This mass represents sloughed mucosa, and is characteristic of gangrenous cholecystitis.

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TABLE 41.1 Antibiotic Penetration into Bile	
Good Penetration	**Poor Penetration**
Ampicillin/Sulbactam	Cefepime
Ceftriaxone	Cefotaxime
Ciprofloxacin	Ceftazidime
Levofloxacin	Gentamicin
Piperacillin/tazobactam	Meropenem
Tigecycline	Vancomycin

Management

Prompt intervention is mandatory, as this condition can deteriorate rapidly. Laparoscopic cholecystectomy is the treatment of choice (2), but for patients who are too unstable for surgery, percutaneous drainage of the gallbladder is a suitable alternative.

Empiric antibiotic therapy should be started as soon as the diagnosis is confirmed, and coverage should include gram-negative enteric organisms. Table 41.1 shows the penetration of different antibiotics into bile.

CLOSTRIDIUM DIFFICILE

As described in Chapter 4, the normal microbial environment in the GI tract, known as the *microbiome*, has an important role in preventing colonization by undesirable pathogens (10). Disruption of the microbiome, which is known as *dysbiosis*, can lead to a variety of infections (both localized and disseminated), and the most prominent of these infections is *Clostridium difficile* enterocolitis, which *is considered the most common healthcare-associated infection* (11). (*Note: Clostridium difficile* has been renamed *Clostridioides difficile*, but the original name will be used here.)

Pathogenesis

Clostridium difficile is a spore-forming gram-positive anaerobic bacillus that is an uncommon inhabitant of the bowel in healthy subjects, but is able to proliferate in the bowel when the normal microbial population has been altered. The major risk factors for *C. difficile* colonization are advanced age, contact with a healthcare facility, and antibiotic therapy (12,13). The organism is occasionally found in poultry, fish, and red meat (14), but the major route of transmission is the fecal-oral route.

C. difficile is not an invasive organism, but releases cytotoxins (Toxins A and B) that damage the bowel mucosa. This leads to inflammatory infiltration of the bowel wall and symptomatic disease. Severe inflammation is accompanied by raised, plaque-like lesions on the mucosal surface known as ‘pseudomembranes’. The presence of these lesions (*pseudomembranous colitis*) is evidence of severe disease.

The enterocolitis from *C. difficile* is primarily a nosocomial infection (i.e., appears after 48 hours of the hospital stay), and the organism is transmitted from patient-to-patient via the hands of hospital personnel. However, there has been a drastic rise in community-acquired *C. difficile* infections in recent years, and they now account for 34–48% of all cases (11).

Gastric Acid Suppression

There are several reports showing that use of acid-suppressing drugs, particularly proton pump inhibitors, is associated with an increased risk of *C. difficile* infection (15–17). This is consistent with the role of gastric acidity as an antibacterial defense mechanism, which is described in Chapter 4 (see Figure 4.3). The protective effect of gastric acid on *C. difficile* infections has important implications for the escalating and excessive use of gastric acid-suppressing drugs, particularly proton pump inhibitors, for stress ulcer prophylaxis in hospitalized patients. (This topic is revisited in the very last section of the chapter.)

Clinical Features

The principal manifestation of *Clostridium difficile* infection (CDI) is a watery diarrhea, which can be accompanied by a low-grade fever (≤101° F) and mild leukocytosis (WBC count <15,000/mm^3). Fulminant CDI occurs in 3–5% of cases (18), and is characterized by hypotension and abdominal distension from ileus, or from the most feared complication of CDI, toxic megacolon. The latter condition, which is shown in Figure 41.2, is a surgical emergency.

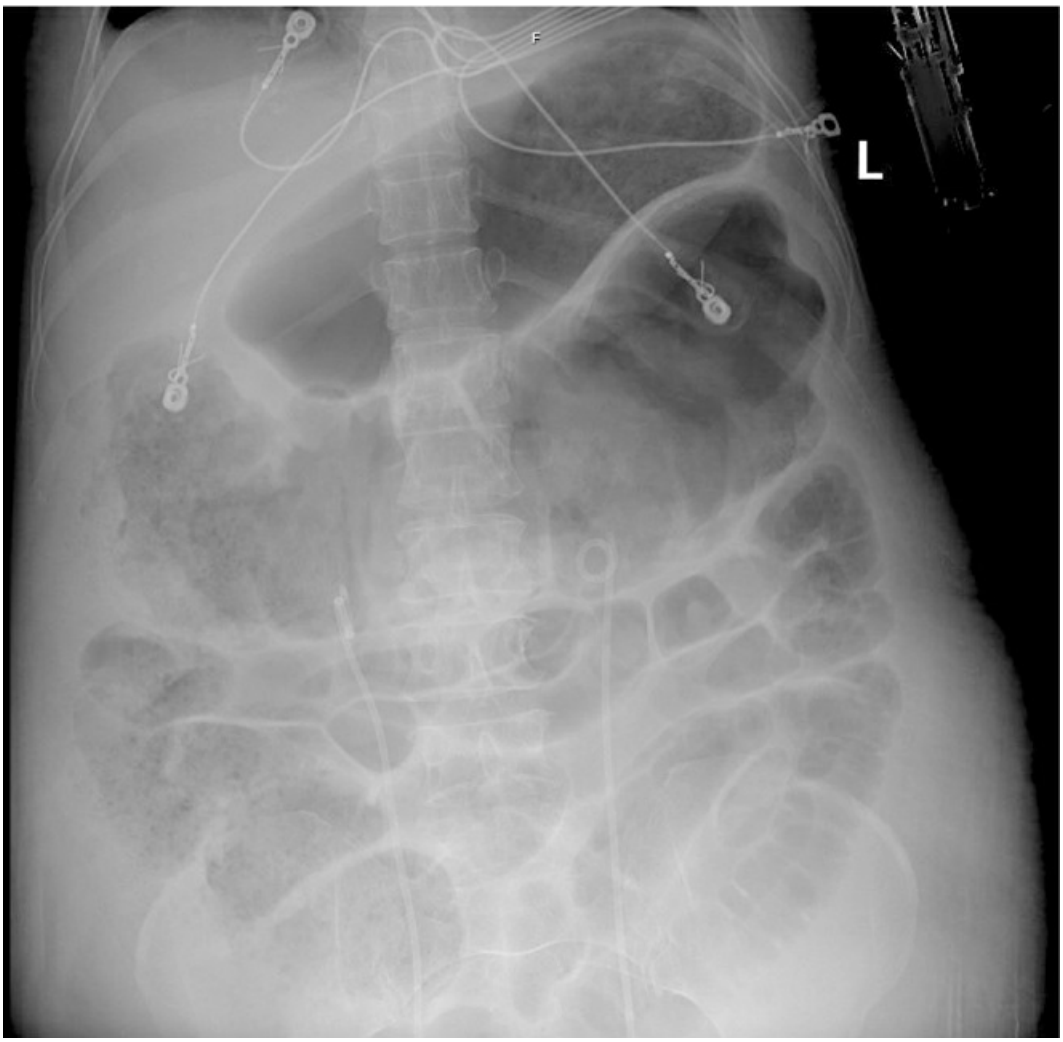


FIGURE 41.2 Radiographic appearance of toxic megacolon in a patient with *C. difficile* enterocolitis.

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Diagnostic Evaluation

The following laboratory tests (on stool samples) are used for the diagnostic evaluation of CDI.

. A polymerase chain reaction (PCR) that detects the gene for Toxin B. A positive test confirms the presence of toxigenic *C. difficile*, but does not differentiate colonization from infection.

. An enzyme-linked immunosorbent assay (ELISA) that detects toxins A and B. A positive test confirms the diagnosis of CDI.

The flow diagram in Figure 41.3 shows how these tests are used in a case of suspected CDI. The PCR is highly sensitive, and is used as an initial screening test. A negative PCR eliminates the diagnosis of CDI (by eliminating the presence of toxigenic *C. difficile*). However, a positive PCR confirms colonization, but not infection, so an ELISA test for cytotoxins is warranted. A positive ELISA test confirms the diagnosis of CDI. However, the ELISA test has a low sensitivity, and can miss about 20% of cases of CDI (12), so a negative ELISA does not eliminate the possibility of CDI. Nevertheless, most patients who are PCR-positive but toxin-negative do well without treatment for CDI (19).

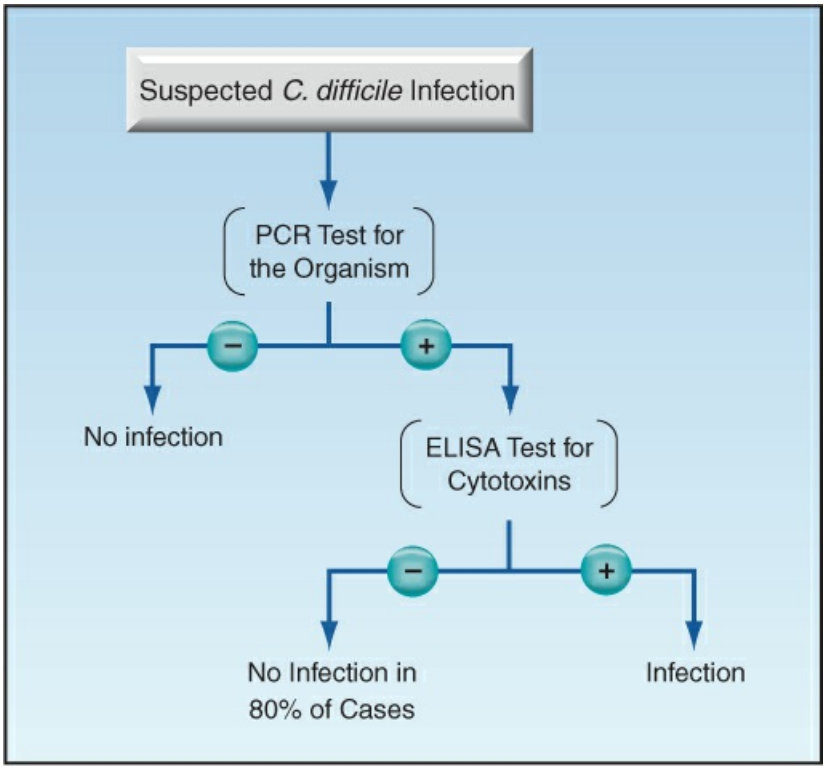


FIGURE 41.3 Flow diagram for the diagnostic evaluation of *C. difficile* infection. See text for explanation.

CAVEAT: There is a tendency to treat patients for CDI based on a positive PCR test only, but about half of patients who are PCR-positive do not have cytotoxins by immunoassay (19), and these patients do well without antibiotics (as just mentioned). Therefore, the treatment of CDI based on the PCR test alone should be discouraged, as it leads to the overdiagnosis of CDI, and the unnecessary use of antibiotics.

COLONOSCOPY: Direct visualization of the bowel mucosa is reserved for the occasional case where there is a high clinical suspicion of CDI that is not confirmed by the cytotoxin assay. The presence of pseudomembranes confirms the diagnosis of CDI. Colonoscopy is preferred to proctosigmoidoscopy for optimal results.

Treatment

The recommended antibiotic regimens for CDI are shown in Table 41.2 (13), and are summarized below.

. The treatment of choice for routine cases of CDI is oral vancomycin or fidaxomicin in the doses shown in Table 41.2. The fever resolves in 24 to 48 hrs, and the diarrhea resolves in 4 to 5 days (20), but treatment is continued for 10 days.

. Fulminant cases of CDI should be treated with a combination of enteral vancomycin and IV metronidazole. If surgery is needed for toxic megacolon, subtotal colectomy is preferred (13).

. Relapses (usually within 3 weeks) are reported in 25% of cases treated with vancomycin (21), and 13% of cases treated with fidaxomicin (22). If vancomycin was used initially, the first recurrence is treated with a 10-day course of fidaxomycin or an extended course of vancomycin, as shown in Table 41.2.

TABLE 41.2 Antibiotic Therapy for *Clostridium difficile* Infection (CDI)

Condition	Recommendations
Initial Episode	Vancomycin: 125 mg PO four times daily for 10 days, *OR* Fidaxomicin: 200 mg PO twice daily for 10 days.
Fulminant CDI	Vancomycin: 500 mg every 6hrs by mouth, NG tube, or rectal enema, *PLUS* Metronidazole: 500 mg IV every 8 hrs.
1st Recurrence	Vancomycin: 125 mg PO four times daily for 10–14 days, then twice daily for 7 days, then daily for 7 days, then every 2–3 days for 2–8 wks, *OR* Fidaxomicin: 200 mg PO twice daily for 10 days. (Only if vancomycin was used for the initial R_x).
2nd Recurrence	Fecal microbiota transplantation

From the clinical practice guidelines in Reference 13.

Fecal Transplantation

About 5% of patients with CDI have more than one relapse (20), and the most effective treatment for multiple relapses is “fecal microbiota transplantation”, which is a fancy term for the instillation of liquid stool from healthy donors (via nasogastric tube, endoscopy, or enemas) to re-establish a normal microbial environment in the GI tract. The reported cure rate is a remarkable 80–100% for recurrent CDI (23).

Preventive Measures

Prophylactic measures are aimed at preventing transmission of the organism from patient to

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patient, and preventing colonization of the organism in susceptible patients.

Infection Control

The guidelines from the Infectious Diseases Society of America recommends gloves and gowns for any contact with an infected patient, and handwashing before and after each patient contact (13). However, these measures are unlikely to eliminate spread of this disease, because asymptomatic carriers of *C. difficile* are important sources of transmitted disease (13).

Probiotics

Probiotics (i.e., non-pathogenic organisms that bind to epithelial cells and prevent the attachment of *C. difficile*) are widely used during and after treatment of CDI (24), but this is not recommended by the American College of Gastroenterology (12) because of inconclusive study results, and evidence that *probiotics can actually impede recolonization of the colon following antibiotic therapy* (25).

COMPLICATED INTRA-ABDOMINAL INFECTIONS

Complicated intra-abdominal infections are defined as infections that involve the peritoneal cavity, and can be a generalized peritonitis or a localized abscess (26). They are often the result of a rupture or tear somewhere along the gastrointestinal tract, or an anastomotic leak following abdominal surgery. These infections can be difficult to treat, and they require prompt source control (i.e., closure of the bowel perforation) and appropriate empiric antibiotics.

Bowel Perforation

Perforations of the GI tract can arise from a wide variety of conditions, including trauma, peptic ulcer disease, bowel ischemia or obstruction, inflammatory or neoplastic disease, rupture of diverticuli, or unintentional laceration of the bowel during surgery. Abdominal pain is almost universal, and there may be signs of peritoneal irritation (guarding and rebound tenderness). Progression to circulatory shock can be rapid (26).

Diagnosis

The diagnosis of bowel perforation requires evidence of free air in the peritoneal cavity (pneumoperitoneum). The presence of free air under the diaphragm in an upright chest x-ray, as shown in Figure 41.4, has a reported sensitivity of up to 85% for the detection of bowel perforations (27). CT scans are more sensitive than plain radiographs, and they can localize the site of the perforation in 85% of cases (28). Pneumoperitoneum is not a useful finding shortly after abdominal surgery, as it can persist for at least one week after the procedure (29).

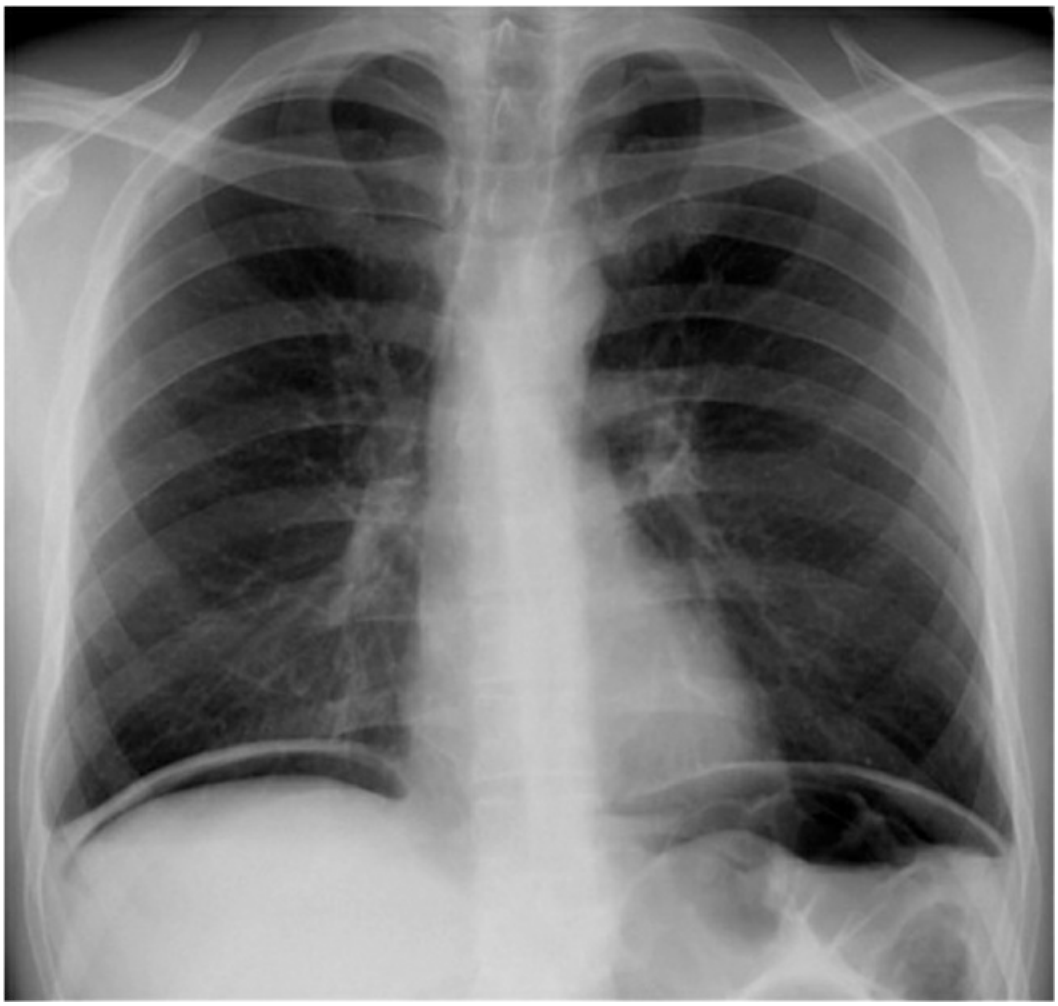


FIGURE 41.4 A chest x-ray in the upright position showing free air under both hemidiaphragms.

Management

Immediate surgical exploration is typically the first priority for bowel perforations, but the perioperative management should include the following measures.

FLUIDS: Soiling with bowel contents can be accompanied by considerable fluid loss into the peritoneal cavity, so signs of circulatory compromise (i.e., a decrease in urine output or blood pressure) should prompt appropriate volume resuscitation. Avoiding vasopressor therapy is optimal prior to surgery (if possible), because it promotes splanchnic vasoconstriction, which will aggravate an underlying ischemic condition in the bowel.

TABLE 41.3 Antibiotic Coverage for Enteric Organisms

A L G r a w a n y

	Gm⁻ Aerobic Bacilli	*Pseudomonas* Species	Enterococci (non-resistant)	Anaerobic Organisms	ESBLs
Piperacillin/ tazobactam	+	+	+	+	+ / –
Meropenem	+	+	+ / –	+	+
Tigecycline	+	–	+	+	+
Ceftazidime/ avibactam	+	+	–	–	+
Amikacin	+	+	–	–	+ / –
Ceftriaxone	+	–	–	–	–
Metronidazole	–	–	–	+	–

ESBLs = extended-spectrum beta-lactamases.

ANTIBIOTICS: Empiric antibiotic therapy should be started as soon as possible using antibiotics that are active against enteric pathogens. The coverage of selected antibiotics is shown in Table 41.3. This is used to make the following recommendations, which are taken from recently published guidelines on intra-abdominal infections (26):

. If no resistant organisms are suspected, *piperacillin-tazobactam* is suitable for single-agent empiric coverage (26). The combination of *ceftriaxone* and *metronidazole* is an alternative recommendation for patients that are not seriously ill, but the coverage is far inferior to piperacillin-tazobactam.

. Organisms that produce extended-spectrum beta-lactamases (ESBLs) are resistant to penicillins and third-generation cephalosporins. If these organisms are suspected (e.g., from prior colonization or infection), the antibiotic options are *meropenem*, *tigecycline*, and *ceftazidime/avibactam*.

a. Meropenem provides the broadest coverage, and is the treatment of choice in patients with septic shock (26). However, there is a reason to limit its use because of rapidly developing resistance (30). Of note, meropenem is not active against methicillin-resistant *Staphylococcus aureus* (MRSA) or vancomycin-resistant enterococci (VRE).
b. Tigecycline does not cover *Pseudomonas* species, but it is active against MRSA and VRE. It is not recommended in patients with septic shock (26).
c. Ceftazidime/avibactam is one of the newer group of beta-lactam antibiotics that contains a beta-lactamase inhibitor (avibactam). Ceftazidime provides coverage for *Pseudomonas*

species, but it is not active against enterococci or anaerobic organisms. The experience with this antibiotic combination is limited at present.

. For patients that are allergic to beta-lactam antibiotics, the combination of amikacin and metronidazole provides suitable coverage.

CANDIDA COVERAGE: *Candida albicans* and related species can be prominent inhabitants of the bowel (especially in patients who have received antibiotics recently), and *perforation in the upper GI tract has been identified as a major risk factor for Candida peritonitis* (31). If this is a concern, empiric coverage should include an antifungal agent that covers all *Candida* species, such as one of the echinocandins (i.e., *micafungin, anidulafungin,* or *caspofungin*).

Postoperative Infections

Postoperative peritonitis is the most common abdominal infection in ICU patients (32), and is usually the result of anastomotic leakage. The management is similar to that described for bowel perforations, with the following exceptions (32):

. The empiric antibiotic regimen should include coverage for ESBL-producing organisms.

. Empiric coverage for *Candida* species is recommended in unstable or critically ill patients.

These infections are problematic because there is often a delay in recognizing an anastomotic leak, which is partly due to the presence of peritoneal air for at least one week after abdominal surgery. Return to the operating room is often prompted by a deterioration in the patient's clinical condition, which increases the risk associated with the reoperation. There is some evidence that C-reactive protein levels can be used to predict the likelihood of an anastomotic leak (33), but this has not gained widespread acceptance.

Abscess

Postoperative abdominal abscesses are often an occult source of sepsis, and are difficult to uncover with routine clinical evaluations. Fever is almost always present, but localized abdominal tenderness can be absent in 60% of cases, and a palpable abdominal mass is evident in less than 10% of cases (34,35). Abdominal x-rays can show extraluminal air, but this occurs in less than 15% of cases (35).

DIAGNOSIS: Computed tomography (CT) of the abdomen is the most reliable method of detecting abdominal abscesses. However, CT imaging in the early postoperative period can be misleading because collections of blood or irrigant solutions in the peritoneal cavity can be misread as an abscess. CT scans are most reliable when performed after the first postoperative week (when peritoneal fluid collections have resorbed) (35). The CT appearance of an abdominal abscess is shown in Figure 41.5.

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FIGURE 41.5 Abdominal CT scan showing a multiloculated abscess in the left upper quadrant in a post-splenectomy patient.

MANAGEMENT: Immediate drainage is advised for postoperative abdominal abscesses. CT-guided drainage is successful in 90–95% of cases (36), and surgery is rarely needed. The empiric antibiotic regimen is the same as described for postoperative peritonitis.

A FINAL WORD

C. difficile and Gastric Acid

The role of gastric acid as an antibacterial defense against ingested microbes (which is described in Chapter 4) does not receive the attention it deserves. One example of this is the evidence that suppression of gastric acidity with proton pump inhibitors promotes the transmission of *Clostridium difficile* infections (15–17), which has hardly curbed the excessive use of proton pump inhibitors for stress ulcer prophylaxis.
The following observation deserves mention:

> There was a marked increase in the incidence of *C. difficile* infections in the first decade of the 21st century, at the same time that proton pump inhibitors gained widespread use for stress ulcer prophylaxis (37).

Is this serendipity, or a causal link? Whatever the answer, the antibacterial actions of gastric acidity are reason enough to curb the excessive use of gastric-acid suppressing drugs for stress ulcer prophylaxis.

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Chapter 42

Urinary Tract Infections in the ICU

Throughout nature, infection without disease is the rule rather than the exception.

René Dubos (*a*)

Indwelling bladder catheters are commonplace in critically ill patients, and catheter-associated urinary tract infections are cited as the most prevalent nosocomial infections in the United States (1). However, this claim may be misleading, since there is a tendency to confuse asymptomatic bacteriuria (i.e., *infection without disease*) with catheter-associated infections. This chapter describes the relevant features of catheter-associated urinary tract infections (UTIs), including pathogenesis, prevention, diagnostic criteria, and treatment.

PATHOGENESIS

The presence of an indwelling bladder catheter is associated with a 3–8% incidence of bacteriuria ($\geq 10^5$ colony forming units/mL) *per day* (2). Bacteria (mostly of bowel origin) form *biofilms* on the inner and outer surface of urethral catheters (3), and these biofilms can serve as a source of continued microbial colonization in the bladder. However, this is not the full story, because *direct injection of pathogens into the bladder of healthy subjects does not result in an infection* (4).

Bacterial Adherence

The missing piece of the puzzle is the ability of pathogenic organisms to adhere to the bladder epithelium. The epithelial cells of the bladder are normally coated with commensal (non-pathogenic) organisms, as shown in Figure 42.1 (5), and this prevents the attachment of pathogenic organisms, which serves as a prelude to urinary tract infections (2,3). This is the same phenomenon that occurs in colonization of the oral mucosa with pathogenic microbes (see Figure 5.5), which is a prelude to nosocomial pneumonia. The presence of a bladder catheter triggers a localized inflammatory response in the bladder epithelium, with exfoliation of the normal epithelial lining, and this is believed to favor the attachment of urinary pathogens (3).

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Microbiology

A multistate survey of the organisms isolated in catheter-associated UTIs (6) included (in order of decreasing prevalence) *Escherichia coli* (28% of cases), *Klebsiella pneumoniae* (23%), enterococci (17%), *Pseudomonas aeruginosa* (10%), *Candida* species (5%), staphylococci (3%) and streptococci (3%). A single organism predominates in short-term catheterization (<30 days), while polymicrobial infections are common in long-term catheterization (≥30 days) (7).

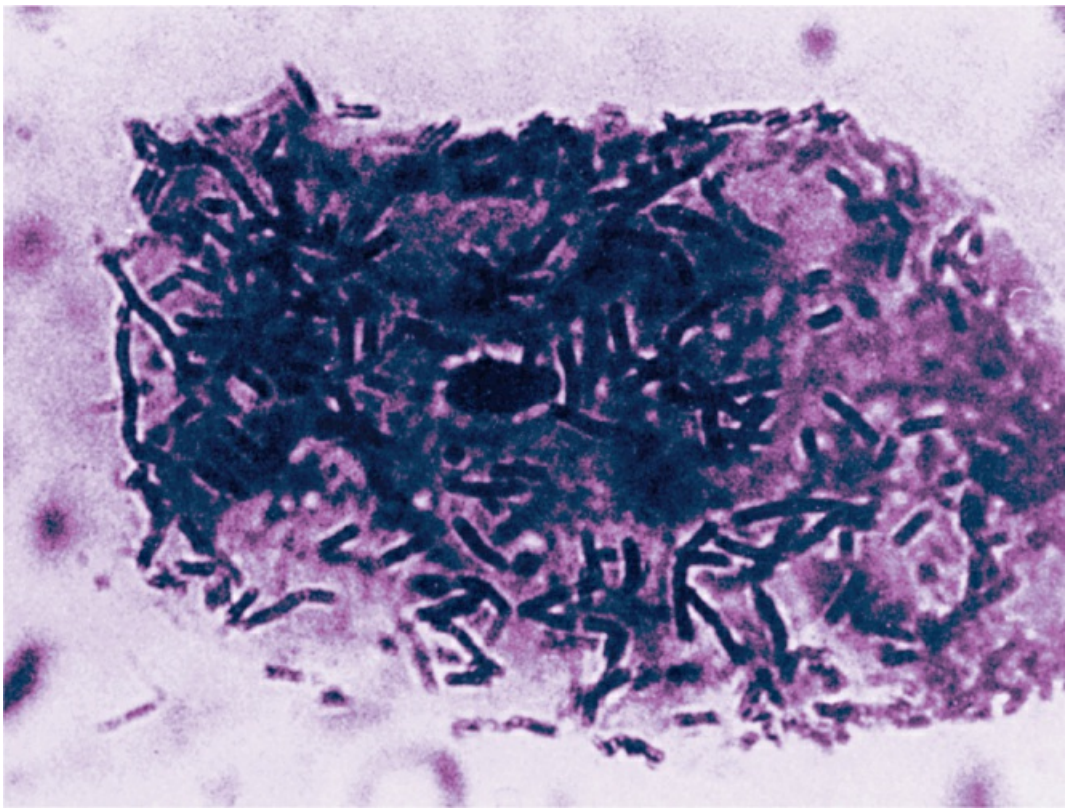


FIGURE 42.1 Photomicrograph showing non-pathogenic *Lactobacillus acidophilus* organisms adhering to a bladder epithelial cell. Image from Reference 5, digitally enhanced.

Prevention

The risk of catheter-associated infection is determined primarily by the duration of catheterization. This is demonstrated in the graphs in Figure 42.2, which show that early removal of urethral catheters reduces the risk of both asymptomatic bacteriuria and catheter-associated infections (8). Removing catheters when they are no longer necessary is the single most effective measure for preventing catheter-associated infections. Other observations about prevention are summarized below.

. Daily cleansing of catheter insertion sites (with antiseptic solutions, antibiotic creams, or soap and water) is not recommended, because this practice can increase the risk of bacteriuria (2).

. Prophylaxis with systemic antibiotics is not recommended (2).

. Antimicrobial-impregnated urinary catheters (i.e., with silver alloy) can reduce the incidence of catheter-associated infections, but only if combined with regular replacement of catheters (9).

DIAGNOSIS AND TREATMENT

The diagnosis of catheter-associated UTIs (CAUTIs) is based on criteria presented by the National Healthcare Safety Network (NHSN) (10), which is a division of the Centers for Disease Control (CDC) that monitors healthcare-associated infections in the United States.

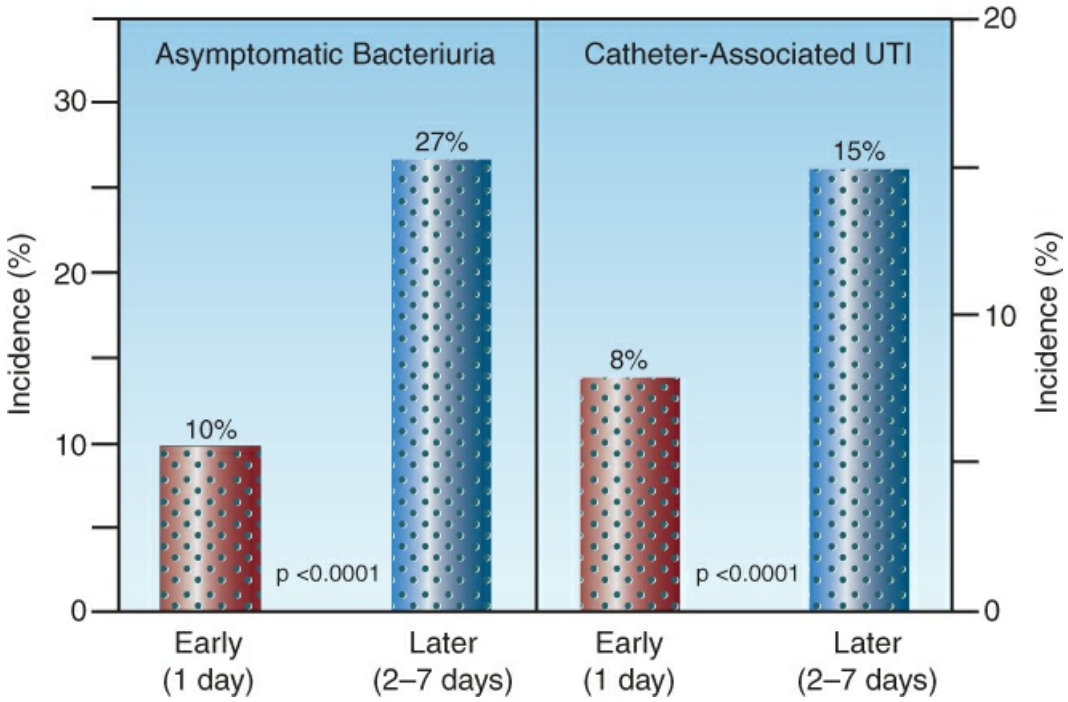


FIGURE 42.2 Graphs showing that the early removal of urethral catheters significantly reduces the incidence of both asymptomatic bacteriuria and catheter-associated infection. Data represents the pooled results of 16 clinical trials, from Reference 8.

Diagnostic Criteria

The diagnosis of CAUTI requires all of the following:

. The presence of an indwelling urethral catheter for at least two consecutive days in an inpatient setting, and the catheter is either still in place, or it has been removed the day before the suspected infection.

. The patient should experience at least one of the following:

a. Fever (>38° C or 100.4° F)
b. Suprapubic tenderness
c. Costovertebral pain or tenderness
d. Urinary urgency or frequency
e. Dysuria

Note: The last two criteria do not apply to patients with indwelling catheters.

. The urine culture should identify no more than two species of organisms, and at least one species should be a bacterium with a growth of $>10^5$ colony-forming units per mL (cfu/mL).

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Fever

The inclusion of fever as a diagnostic criterion for CAUTI is problematic, because there are a multitude of potential causes of fever in ICU patients (see Chapter 44). The nonspecific aspect of fever is demonstrated by a comparison study of patients with and without CAUTI, where the incidence of fever was the same in both groups of patients (11).

Pyuria

Pyuria (i.e., >10 white blood cells per high-powered field in centrifuged urine) is not diagnostic of CAUTI, since there are noninfectious inflammatory conditions that can cause pyuria. However, *the absence of pyuria is evidence against the diagnosis of CAUTI* (2).

Bacteremia

About 20% of bacteremias in hospitalized patients are the result of CAUTI (8), but bacteremia is not included as a diagnostic criterion for CAUTI. However, the NHSN has a diagnostic entity called *Asymptomatic Bacteremic Urinary Tract Infection* (ABUTI), which is a patient with or without a urethral catheter that is asymptomatic but has significant bacteriuria ($>10^5$ cfu/mL) with an organism that is also growing in blood cultures. In other words, *if the same bacterium is isolated in blood and urine cultures (and growth in the urine culture is $>10^5$ cfu/mL) the diagnosis is a UTI, regardless of the presence or absence of symptoms*. This does not apply to *Candida* species, as described later.

Asymptomatic Bacteriuria

The diagnosis of asymptomatic bacteriuria thus requires the following:

. A urine culture that shows growth of a single bacterium with a colony count $>10^5$ cfu/mL.

. Blood cultures that show no growth of the bacterium identified in the urine culture.

. The absence of symptoms related to a UTI.

Treatment

. Antibiotic treatment of patients with asymptomatic bacteriuria is not advised unless the patient is scheduled for a urologic procedure that results in mucosal bleeding (12).

. Empiric antibiotics are recommended for patients with suspected CAUTI, and should provide coverage for gram-negative aerobic bacilli and enterococci.

a. Single agent therapy with *piperacillin-tazobactam* or a carbapenem (e.g., *meropenem*) is suitable (13).

b. Alternative (second-line) agents include *ceftazidime*, *cefepime*, or a fluoroquinolone (e.g., *levofloxacin*) (13).

. If the diagnosis of CAUTI is confirmed by urine culture, antibiotic therapy should be tailored to the isolated organism and microbial sensitivities. Catheters that have been in place for longer than two weeks should be replaced or removed if no longer necessary (2).

. The duration of antibiotic therapy for CAUTI should be 7 days for patients who respond promptly, and 10–14 days otherwise (2,13).

Candiduria

The presence of *Candida* species in urine usually represents colonization in patients with indwelling urethral catheters, but candiduria can also be a sign of disseminated candidiasis (i.e., the candiduria being the result, not the cause, of the disseminated candidiasis). In fact, *candiduria may be the only evidence of disseminated disease*, since blood cultures are unrevealing in more than 50% of cases of disseminated candidiasis (14).

Microbiology

In cases of candiduria, the colony count has no predictive value for identifying renal or disseminated candidiasis (14). The most frequent isolate is *C. albicans* (about 50% of cases), followed by *C. glabrata* (14,15). The latter organism is notable for resistance to the antifungal agent fluconazole.

Asymptomatic Candiduria

Asymptomatic candiduria does not require treatment unless the patient is neutropenic (and is at risk for disseminated candidiasis) (15,16); in this case, the recommended prophylaxis is an echinocandin (i.e., *caspofungin, micafungin,* or *anidulafungin*) (16).

Symptomatic Candiduria

Candiduria that is associated with fever, suprapubic tenderness, or costovertebral tenderness requires antifungal therapy as well as blood cultures and imaging studies of the kidneys (with ultrasound or computed tomography) to search for renal abscesses or evidence of urinary tract obstruction.

. If systemic candidiasis is suspected, or is verified by positive blood cultures, the recommended treatment is an echinocandin. These agents are preferred to fluconazole because they are active against all *Candida* species.

. The recommended treatment for *Candida* UTIs (cystitis or pyelonephritis) is dependent on the organism (16).

a. For *Candida* species that are sensitive to fluconazole (e.g., *Candida albicans*), the treatment is *oral fluconazole* (200–400 mg daily for 2 weeks) (17).
b. For fluconazole-resistant organisms (i.e., *Candida glabrata* and *Candida krusei*), the recommended treatment is *amphotericin B* (0.3–0.6 mg/kg daily for 1–7 days). The echinocandins are not recommended in this situation because of poor penetration into the urine.

. For *Candida* UTIs associated with fungus balls, surgical intervention is recommended. If nephrostomy tubes are present, irrigation with *amphotericin B* (25–50 mg in 200–500 mL sterile water) is recommended.

A FINAL WORD

Bacterial Adherence & Nosocomial Infections

The unifying feature in nosocomial infections that involve the gastrointestinal, respiratory, and urinary tracts is a change in the character of microbes that adhere to epithelial surfaces. In healthy subjects, the epithelial surfaces in the mouth, GI tract, and urinary tract are colonized by harmless microbes, but this changes in patients who are seriously ill, when the epithelial surfaces become populated with pathogenic organisms. This pathogenic colonization then serves as a prelude to nosocomial infections.

The colonization of epithelial surfaces is not a function of microbial density (where the most densely populated organisms take up the space on the epithelial surface), but rather is determined by the ability of bacteria to bind to the epithelial cells. For example, pathogenic *E. coli* (the most common cause of CAUTIs) have specialized appendages called *pili* that can bind to glycoproteins on bladder epithelial cells, and there are small carbohydrate molecules (such as mannose) that can competitively block this binding (3). Observations like this suggest a novel approach to the prevention of nosocomial infections by preventing the adhesion of pathogenic organisms to epithelial surfaces. This approach is appealing because it does not require prophylactic antibiotics, and thus would limit the emergence of resistant organisms.

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Section XIII

ALTERED BODY TEMPERATURE

There is no possibility of escaping the entropic doom imposed on all natural phenomena.

Aharon Katchalsky
1965

Chapter 43

Thermoregulatory Disorders

Heat not a furnace for your foe so hot that you singe yourself.

Shakespeare
Henry VIII

The human body is a metabolic furnace that generates enough heat to raise the body temperature by 1° C every hour, even at rest (1). Fortunately, the external surface of the body acts like a radiator, and discharges excess heat into the surrounding environment. The behavior of this radiator is guided by a thermostat (the thermoregulatory system) that limits the daily variation in body temperature to ±0.6° C (2). This chapter describes what happens when this thermostat fails, and allows the body temperature to rise or fall to life-threatening levels.

HEAT-RELATED ILLNESS

Hyperthermia vs. Fever

The distinction between *hyperthermia* and *fever* deserves mention at the outset. Both conditions are characterized by an elevated body temperature, but hyperthermia is the result of a defect in temperature regulation, while fever is the result of a normal thermoregulatory system operating at a higher set point. The elevations in body temperature in this chapter represent hyperthermia, not fever. Because the underlying mechanisms involved in the production of hyperthermia and fever are different, *the antipyretic agents used to treat fever (e.g., acetaminophen) are ineffective in hyperthermia*.

Response to Thermal Stress

The maintenance of body temperature in conditions of thermal stress (e.g., hot weather, strenuous exercise) is primarily achieved by enhanced blood flow to the skin (convective heat loss) and the loss of sweat (evaporative heat loss).

Convective Heat Loss

When heat is lost from the skin, it warms the air just above the skin surface, and the increase in

surface temperature limits the further loss of body heat by conduction. However, when an air current (e.g., from a fan or gust of wind) is passed across the skin, it displaces the warm layer of air above the skin and replaces it with cooler air, and this process facilitates the continued loss of body heat by conduction. The same effect is produced by increases in blood flow just underneath the skin. The action of currents (air and blood) that promote heat loss is known as *convection*.

Evaporative Heat Loss

The transformation of water from a liquid to a gas requires heat (called the 'latent heat of vaporization'), and the heat required for the evaporation of sweat from the skin is provided by body heat. The evaporation of one liter of sweat from the skin is accompanied by the loss of 580 kilocalories (kcal) of heat from the body (3). This is about one-quarter of the daily heat production by an average-sized adult at rest. Thermal sweating (as opposed to "nervous sweating") can achieve rates of 1 to 2 liters per hour (3), which means that over 1,000 kcal of heat can be lost in one hour during profuse sweating. It is important to emphasize that *sweat must evaporate to ensure loss of body heat. Wiping sweat off the skin will not result in heat loss*, so this practice should be discouraged during strenuous exercise.

Syndromes

Heat-related illnesses are conditions where the thermoregulatory system is no longer able to maintain a constant body temperature in response to thermal stress. There are a number of minor heat-related conditions, such as *heat edema* (dependent soft tissue swelling), *heat cramps* (exercised-induced muscle cramps), and *heat rash* (a papular rash caused by blocked sweat pores) (4), but the following descriptions are limited to the major heat-related illnesses: *heat exhaustion* and *heat stroke*. The comparative features of these conditions are shown in Table 43.1 (3–5).

TABLE 43.1 Comparative Features of Heat Exhaustion and Heat Stroke

Feature	Heat Exhaustion	Heat Stroke
Body Temperature	38–39° C or 100.4–102° F	≥41° C or ≥106° F
CNS Dysfunction	No	Yes
Sweat Production	Yes	No
Dehydration	Yes	Yes
Hemodynamic Instability	No	Yes
Multiorgan Involvement	No	Yes

Heat Exhaustion

Patients with heat exhaustion experience flu-like symptoms that include hyperthermia (usually <38–39° C or 100.4–102° F), muscle cramps, nausea and malaise. The hallmark of this condition is *dehydration without signs of hemodynamic compromise*. The volume loss can be accompanied by hypernatremia (from sweat loss) or hyponatremia (when sweat loss is replaced with water

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intake). Mentation is usually intact.

The management of heat exhaustion includes volume repletion and other general supportive measures (e.g., placing the subject in an air conditioned room). Active cooling measures are not necessary.

Heat Stroke

Heat stroke is a life-threatening condition characterized by extreme elevations in body temperature (≥41° C [≥106° F]), severe neurologic dysfunction (e.g., delirium, coma, and seizures), severe volume depletion with hypotension, and multiorgan involvement that includes rhabdomyolysis, acute kidney injury, disseminated intravascular coagulopathy (DIC), and marked elevation in serum transaminases, presumably from "ischemic hepatitis". The inability to produce sweat (anhidrosis) is a typical, but not universal, feature of heat stroke (5).

There are two types of heat stroke: (*a*) *classic heat stroke*, which is related to environmental temperatures, and (*b*) *exertional heat stroke*, which is related to strenuous exercise. Exertional heat stroke tends to be more severe, with a higher incidence of multiorgan dysfunction.

Management

The management of heat stroke includes volume resuscitation and active cooling measures to reduce the body temperature to 38° C (100.4° F).

EXTERNAL COOLING: External cooling in the field is best achieved by immersion in cold water or ice water. If these are not available, *evaporative cooling* is effective. This involves spraying the skin with cool water and then fanning the skin to promote evaporation. This method can reduce the body temperature at a rate of 0.3° C (0.6° F) per minute (6). Evaporative cooling is most effective when the weather is hot and dry (which enhances evaporation from the skin).

External cooling in the emergency department is accomplished by placing ice packs in the groin and axilla, and covering the upper thorax and neck with ice (7). Cooling blankets are then placed over the entire length of the body.

The major drawback of external cooling is the risk of shivering, which is counter-productive because it raises the body temperature.

INTERNAL COOLING: Internal cooling can be achieved with intravenous infusion of cold saline (4° C). Faster cooling has been reported with automated intravascular cooling devices (8) (i.e., the ones used for targeted temperature management in comatose survivors of cardiac arrest), but this approach requires central venous access, and the impact on outcomes has not been adequately studied.

Rhabdomyolysis

Skeletal muscle injury (rhabdomyolysis) is a common complication of hyperthermia syndromes, including heat stroke and the drug-induced hyperthermia syndromes (described later in the chapter). Disruption of myocytes in skeletal muscle leads to the release of creatine kinase (CK) into the bloodstream, and the measurement of CK levels in plasma is used to determine the presence and severity of rhabdomyolysis. There is no standard plasma CK level for the diagnosis of rhabdomyolysis, but levels that are at least five times normal have been suggested (9).

The adverse consequence of rhabdomyolysis are muscle weakness and acute kidney injury. The latter condition is the result of renal tubular injury from the myoglobin released by disrupted myocytes (10). The acute kidney injury from rhabdomyolysis is described in Chapter 34, along with the management of rhabdomyolysis.

DRUG-INDUCED HYPERTHERMIA SYNDROMES

Hyperthermia can also be a drug-related disorder, and three drug-related hyperthermia syndromes are recognized:

. Malignant Hyperthermia

. Neuroleptic Malignant Syndrome

. Serotonin Syndrome

Each of these is briefly described in this section.

Malignant Hyperthermia

The relevant features of malignant hyperthermia (MH) are summarized in Table 43.2. This condition is a rare inherited disorder with an autosomal dominant pattern that affects about 1 in 50,000 adults (11). It is characterized by excessive release of calcium from the sarcoplasmic reticulum in skeletal muscle in response to halogenated inhalational anesthetic agents (e.g., halothane, isoflurane) and depolarizing neuromuscular blockers (e.g., succinylcholine) (11).

TABLE 43.2 Malignant Hyperthermia

Predisposing Conditions:

A susceptible patient who has received:

1. A halogenated volatile anesthetic agent (e.g., halothane) or
2. A depolarizing neuromuscular blocker (e.g., succinylcholine)

Clinical Presentation:

1. Muscle rigidity
2. Increase in temp to ≥40° C
3. Altered mentation
4. Autonomic instability

Management:†

1. Dantrolene:
   a. 1–2 mg/kg as a rapid IV bolus, and repeat every few minutes until symptoms resolve or the cumulative dose reaches 10 mg/kg.
   b. Follow with oral or IV doses of 1–2 mg/kg every 6 hours for at least 24 hrs (to prevent recurrence).
2. Aggressive fluid infusion may be necessary for autonomic instability or rhabdomyolysis.

†Advice for acute management is available on the Malignant Hyperthermia Hotline: 1-800-644-9737. Dantrolene dosing regimen from Reference 13.

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Clinical Manifestations

The first sign of MH may be a sudden and unexpected rise in end-tidal PCO_2 (11,12), which is a reflection of the increased metabolic rate caused by calcium-induced uncoupling of oxidative phosphorylation. This is followed (within minutes to a few hours) by muscle rigidity, which can begin in the masseter muscle before it becomes generalized. The heat generated by the combination of the hypermetabolism and muscle rigidity results in a marked rise in body temperature (often >40° C or 104° F), which generally occurs later in the presentation (12).

Neurologic dysfunction is commonplace in MH, with altered mentation (e.g., confusion, delirium, coma) and autonomic instability (e.g., fluctuating blood pressure, tachycardia). Rhabdomyolysis is also common, and can be accompanied by acute kidney injury from myoglobinuria.

Management

The first suspicion of MH should prompt immediate discontinuation of the offending anesthetic agent.

DANTROLENE: Specific treatment for the muscle rigidity is available with *dantrolene* sodium, a muscle relaxant that blocks the release of calcium from the sarcoplasmic reticulum. When given early in the course of MH, dantrolene can reduce the mortality rate from ≥70% (in untreated cases) to ≤10% (11,12). The dosing regimen for dantrolene is shown in Table 43.2 (13).

The most common side effect of dantrolene is muscle weakness, particularly grip strength, which usually resolves in 2 to 4 days after the drug is discontinued (13). The most troublesome side effect is hepatocellular injury, which is a risk with oral dantrolene, but requires months of treatment (13).

OTHER CONCERNS: Aggressive fluid infusion may be necessary for autonomic instability or rhabdomyolysis. If rhabdomyolysis develops, vigilance for hyperkalemia and acute kidney injury is necessary. Control of the rigidity should reduce the body temperature, and active cooling measures are usually not necessary.

Note: There is a Malignant Hyperthermia Hotline (1-800-644-9737) that provides advice for the acute management of MH.

Followup

All patients who survive an episode of MH should be given a medical bracelet that identifies their susceptibility to MH. In addition, because MH is a genetic disorder with an autosomal dominant inheritance pattern, immediate family members should be informed of their possible susceptibility to MH. (For testing options, see the website for the Malignant Hyperthermia Association of the United States – mhaus.org.)

Neuroleptic Malignant Syndrome

The neuroleptic malignant syndrome (NMS) is similar to MH in that it is a drug-induced disorder characterized by hyperthermia, muscle rigidity, and altered mentation (14,15).

Pathogenesis

NMS is the result of a decrease in dopamine-mediated transmission in the basal ganglia and hypothalamic-pituitary axis (14). As indicated in Table 43.3, NMS can be caused by drugs that inhibit dopaminergic transmission (most cases), or it can be triggered by discontinuing drugs that *facilitate* dopaminergic transmission. The drug most frequently implicated in NMS is *haloperidol* (14).

TABLE 43.3 Drugs Associated with Neuroleptic Malignant Syndrome

Drugs that Inhibit Dopaminergic Transmission	
Antipsychotic agents:	Butyrophenones (e.g., haloperidol), phenothiazines, clozapine, olanzapine, respiradone
Antiemetic agents:	Metoclopramide, droperidol, prochlorperazine
CNS stimulants:	Amphetamines, cocaine
Other:	Lithium, tricyclic antidepressants
Drugs that Facilitate Dopaminergic Transmission†	
Dopaminergic drugs:	Amantidine, bromocriptine, levodopa

†Discontinuing these drugs can trigger the neuroleptic malignant syndrome.

Despite the association with certain drugs, NMS appears in fewer than 1% of patients treated with high-risk drugs (15), and there is no relationship between drug doses and the appearance of NMS (14). It is thus an idiosyncratic reaction, rather than a dose-dependent toxic reaction.

Clinical Features

Most cases of NMS begin to appear 24 to 72 hours after the initiation of drug therapy, and almost all cases are apparent in the first 2 weeks of drug therapy. The onset is usually gradual, and the initial manifestations are muscle rigidity and agitation or confusion (14). The increase in body temperature can be delayed for 8 to 10 hours after the onset of muscle rigidity (16). Autonomic hyperreactivity is also common, and can produce tachycardia and diaphoresis (15). Severe cases are characterized by *lead-pipe rigidity*, rhabdomyolysis, and depressed consciousness.

Dystonic reactions, which are common complications of neuroleptic agents, can be difficult to distinguish from NMS, especially in the early stage of NMS, when muscle rigidity may be the only manifestation. The plasma level of creatine kinase (CK) can help in this regard, because plasma CK levels are only mildly elevated in dystonic reactions, but are typically higher than 1,000 Units/L in NMS (17).

Management

The single most important therapeutic measure is *immediate* removal of the offending drug. If NMS is caused by discontinuation of dopaminergic drugs, the drug should be restarted immediately, with gradual reduction of the drug dosage at a later time. General measures for NMS can include the following (15):

. Aggressive fluid infusion may be necessary for patients with rhabdomyolysis or hypotension.

. External cooling may be needed to reduce the body temperature, but control of the muscle

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rigidity usually corrects the problem (15).

. Prophylaxis for venous thromboembolism is mandatory, because there is an increased risk of deep vein thrombosis in NMS (14).

. Benzodiazepines are often used in patients who are agitated (15,17).

Management of the muscle rigidity is possible with the following drugs:

BROMOCRIPTINE: Bromocriptine mesylate is a dopamine agonist that has been successful in treating NMS when given orally in a dose of 10 mg three times daily (15). Some improvement in muscle rigidity can be seen within hours after the start of therapy, but the *full response often takes days to develop* (18). Hypotension and hallucinations are troublesome side effects (15).

DANTROLENE: Dantrolene (the same muscle relaxant used in the treatment of MH) can be used for severe cases of NMS (when a more rapid response is desirable), although the success in clinical trials has been inconsistent (15). The optimal dosing regimen is not clearly defined, but one suggested regimen is as follows (15):

. Start with IV dantrolene in a bolus dose of 1 mg/kg every 8 hours, until symptoms resolve, or the cumulative dose reaches 10 mg/kg.

. Follow the initial regimen with oral dantrolene in a daily dose of 50–200 mg given in 3–4 divided doses.

Treatment of NMS should continue for about 10 days after clinical resolution because of delayed clearance of many neuroleptics. When depot preparations are implicated, therapy should continue for 2 to 3 weeks after clinical resolution (14).

Serotonin Syndrome

Overstimulation of serotonin receptors in the central nervous systems produces a variety of neurologic abnormalities that make up what is known as the *serotonin syndrome* (19). This syndrome has received much more attention in recent years, thanks to the popularity of serotonin reuptake inhibitors (SSRIs) for the treatment of depression.

TABLE 43.4 Drugs that can Produce the Serotonin Syndrome

Mechanism	Drugs†
Increased serotonin synthesis	L-tryptophan
Decreased serotonin breakdown	MAO inhibitors (including linezolid), ritonavir
Increased serotonin release	Amphetamines, MDMA (ecstasy), cocaine
Decreased serotonin reuptake	SSRIs, TCAs, dextromethorphan, meperidine, fentanyl, tramadol
Serotonin receptor agonists	Lithium, sumitriptan, buspirone, LSD

†See Reference 19 for a comprehensive list of drugs. Abbreviations: MAO = monoamine oxidase, MDMA = methylene dioxy-methamphetamine, SSRIs = selective serotonin reuptake inhibitors, TCAs = tricyclic

antidepressants.

Pathogenesis

Serotonin is a neurotransmitter that participates in sleep-wakefulness cycles, mood, and thermoregulation. A variety of drugs can enhance serotonin neurotransmission and thereby promote the serotonin syndrome, and a list of these drugs is shown in Table 43.4. Many of these drugs are mood enhancers, including illegal substances like "ecstasy", an amphetamine derivative implicated in life-threatening cases of SS (20). Note also that fentanyl and linezolid (two drugs used in the care of ICU patients) are capable of causing NMS. The serotonin syndrome occurs most often when two serotonergic drugs are used together.

Clinical Manifestations

The clinical features of serotonin syndrome (SS) can be placed in three categories: motor hyperactivity, adrenergic hyperactivity, and altered mentation. This is shown in Table 43.5, along with the frequency of clinical manifestations in each category (21). The following points about the clinical presentation of SS deserve mention:

. The clinical features that are most specific for SS are hyperreflexia, myoclonus, and clonus (spontaneous or inducible) (21,22).

. Although infrequent, *ocular clonus (spontaneous horizontal eye movements) is considered the most reliable clinical finding for the diagnosis of SS* (22).

. Mild cases of SS are often characterized by hyperreflexia, tachycardia, and agitation. Muscle rigidity appears in more severe cases, and can mask the more specific manifestations of SS (e.g., hyperreflexia).

. Although hyperthermia is considered one of the hallmarks of SS, elevated body temperatures are reported in only 60% of cases (21).

. Rhabdomyolysis is reported in 15% of cases of SS (21), and usually appears along with muscle rigidity.

TABLE 43.5 Clinical Manifestations of the Serotonin Syndrome

Category	Manifestations
Motor Hyperactivity	Hyperreflexia (57%) Myoclonus (42%) Clonus (34%) Ocular clonus (7%) } * Rigidity (45%)
Adrenergic Hyperactivity	Increased Temp (60%) Tachycardia (85%) Hyper/hypotension (76%) Mydriasis (34%)

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Altered Mentation	Agitation (56%) Confusion (64%) Coma (11%)

The frequency of clinical manifestation (in parentheses) are from Reference 21. (*) Clinical findings that are the most specific for serotonin syndrome.

SS VERSUS NMS: Many of the clinical features of SS are similar to NMS: i.e., both are characterized by muscle rigidity, altered mental status, increased body temperature, and adrenergic hyperactivity. The distinguishing features of SS are the hyperreflexia and clonus, but severe cases of SS (where muscle rigidity predominates and can mask the hyperreflexia and clonus) can be difficult to distinguish from NMS (15). A careful drug history is often helpful, although these two conditions share some of the causative agents.

Management

As in other drug-induced hyperthermia syndromes, removal of the precipitating drug(s) is the single most important therapeutic measure in SS. The general supportive measures in SS are the same as described for NMS. Most cases of SS resolve within days with general supportive care. Severe cases of SS can benefit from a serotonin antagonist like cyproheptadine.

CYPROHEPTADINE: Cyproheptadine is a serotonin antagonist that is available for oral administration only, but tablets can be broken up and administered through a nasogastric tube. The initial dose is 12 mg, followed by 2 mg every 2 hrs for persistent symptoms, and the maintenance dose is 8 mg every 6 hours (15). This drug is sedating, but that can benefit the agitation in SS.

PARALYSIS: Severe rigidity and rhabdomyolysis that does not improve with conventional measures may require neuromuscular paralysis with a nondepolarizing agent (e.g., rocuronium).

HYPOTHERMIA

Hypothermia is defined as a body temperature <35° C (<95° F), and can be either primary (i.e., the result of environmental conditions), secondary (i.e., triggered by illness or trauma), or therapeutic (i.e., targeted temperature management). The focus of this section is primary (environmental) hypothermia.

Pathogenesis

Physiologically, the human body is better equipped to survive in hot rather than cold environments. The physiological response to cold includes *cutaneous vasoconstriction* (to reduce convective heat loss) and *shivering* (which can double metabolic heat production), but these physiological responses provide protection only in the early stages of hypothermia. Instead, the behavioral responses to cold (e.g., wearing warm clothing or seeking shelter from the cold) play a major role in the adaptation to cold.

Predisposing Conditions

Environmental hypothermia is most likely to occur in the following situations:

. Immersion in cold water (<15° C or <59° F) can be fatal after 30 minutes (23), because heat loss to cold water occurs much more rapidly than heat loss to cold air. This is also why exposure to cold in wet clothing predisposes to hypothermia.

. Windy conditions promote hypothermia in cold environments, because wind promotes heat loss by convection (as described earlier). This is the basis for the wind-chill index, which describes the combined effect of ambient temperature and wind speed on the temperature at the skin surface (24).

. Hpothermia is a risk when the behavioral responses to cold are impaired; e.g., by alcohol or drug intoxication.

Clinical Presentation

The clinical manifestations in progressive hypothermia are summarized in Table 43.6 (23,25).

TABLE 43.6 Manifestations of Progressive Hypothermia

Severity	Body Temp	Clinical Manifestations
Mild	32–35° C 90–95° F	Confusion, cold and pale skin, shivering, tachycardia
Moderate	28–31.9° C 82–89.9° F	Lethargy, reduced or absent shivering, bradycardia, bradypnea
Severe	<28° C <82° F	Obtundation or coma, no shivering, dilated and fixed pupils, bradycardia, hypotension, oliguria, edema
Fatal	<24° C <75° F	Apnea, asystole

Mild Hypothermia

Mild hypothermia is defined as a body temperature of 32–35° C (90–95° F). At this stage, patients are often confused, but they shows signs of adaptation to cold; i.e., the skin is pale and cold from cutaneous vasoconstriction, and shivering is brisk. Tachycardia also predominates at this stage, and the blood pressure can be elevated (26).

Moderate Hypothermia

Patients with moderate hyperthermia (i.e., 28–31.9° C or 82–89.9° F) are lethargic, and shivering may be present if the body temperature is above 30° C (23). Bradycardia and a decreased respiratory rate (bradypnea) predominate at temperatures below 32° C (26).

Severe Hypothermia

Patients with severe hypothermia (<28° C or <82° F) are usually obtunded or comatose, and have dilated, fixed pupils. Additional findings include hypotension, severe bradycardia, oliguria, and generalized edema.

Apnea and asystole are expected at body temperatures below 24° C (75° F), but *the diagnosis*

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of death cannot be made in the setting of hypothermia (25,26). The body must be warmed to a normal temperature before the pronouncement of death. (See also the brain death determination in Chapter 45).

Diagnostic Evaluation

Exposure to a cold environment results in skin temperatures that are lower than interior (core) temperatures, so peripheral temperature measurements (i.e., oral, axillary, temporal artery, and tympanic temperatures) are not appropriate in hypothermic patients (27). Core temperature measurements are advised, using bladder, esophageal, or rectal probes.

Laboratory Evaluation

The following points deserve mention concerning the laboratory evaluation of hypothermic patients:

. Arterial blood gas measurements (which are performed at 37° C) should be adjusted to the patient's body temperature (26). In moderate-to-severe hypothermia, expect a combined respiratory and metabolic acidosis.

. Serum electrolytes can reveal hyperkalemia, from potassium release by skeletal muscle, as a result of shivering and/or rhabdomyolysis.

. Serum creatinine levels can be elevated as a result of rhabdomyolysis, acute renal failure, or *cold diuresis* (caused by diminished renal tubular responsiveness to antidiuretic hormone).

. A generalized coagulopathy (i.e., thrombocytopenia, and prolongation of the prothrombin time and partial thromboplastin times) is common at temperatures below 34° C (93° F) (26), but may not be evident if the coagulation profile is run at normal body temperatures (28).

Electrocardiogram

The electrocardiogram in hypothermia can show positive J (Osborn) waves at the junction between the QRS complex and the ST segment (see Figure 43.1), especially in leads II and V6 (26). These *Osborn waves* are not specific for hypothermia, and can occur with hypercalcemia, subarachnoid hemorrhage, cerebral injuries, and myocardial ischemia (29). Despite the attention these waves have received, they are merely a curiosity, and have little or no diagnostic or prognostic value in hypothermia.

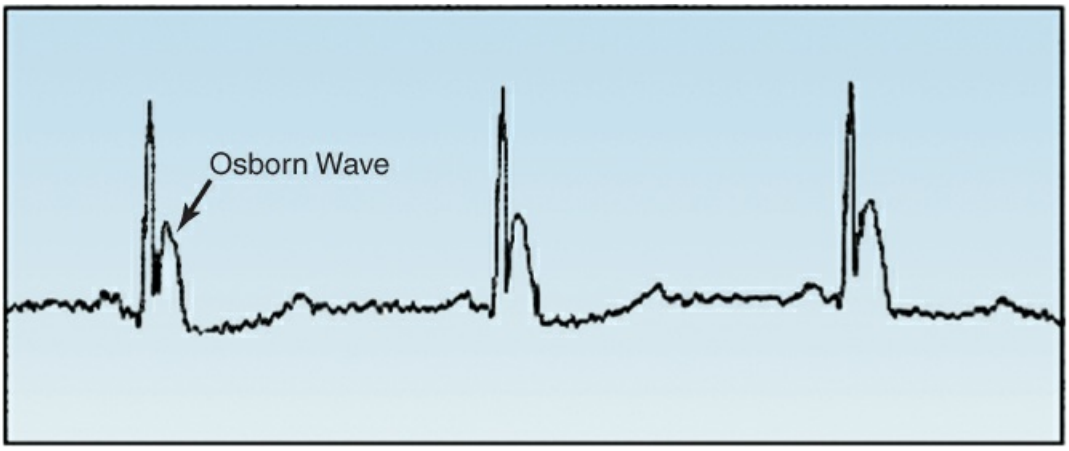


FIGURE 43.1 The (overhyped) Osborn wave.

ARRHYTHMIAS: The threshold for atrial and ventricular arrhythmias is reduced at temperatures below 32° C (90° F), and there is also a progressive bradycardia at temperatures <32° C that is resistant to atropine. Finally, there is a risk of ventricular fibrillation and asystole at temperatures below 28° C (82.4° F) (26).

Rewarming

There are several approaches to rewarming, depending on the severity of the hypothermia.

External Rewarming

External rewarming is adequate for most cases of hypothermia Passive rewarming (e.g., removing wet clothes, covering the patient in blankets) prevents heat loss, and allows the patient's body to rewarm physiologically. Active warming (e.g., with heating pads or automated warming systems) prevents heat loss and accelerates warming.

Internal Rewarming

Internal rewarming is generally reserved for severe cases of hypothermia where active external warming is ineffective. The following are some relevant points about internal rewarming:

. For intubated patients, heating the inhaled gases to 40–45° C (104–113° F) can raise the core temperature 2.5° C per hour (28).

. Heated IV fluids prevent further cooling, but they do not promote rewarming (23).

. Internal rewarming methods that are not recommended include gastric lavage with heated fluid (risk of aspiration), and warmed bladder lavage (ineffective) (23).

. For hypothermic cardiac arrest, venoarterial ECMO has been used successfully (30).

Volume Resuscitation

Patients with moderate-to-severe hypothermia are often volume depleted (from cold diuresis), and the vasodilation that accompanies rewarming promotes hypotension. Therefore, attention to fluid infusion is needed during the rewarming process. However, the infusion of fluids at room temperature (21° C or 70° F) promotes further cooling, so the infused fluids should be heated.

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Patients who are refractory to volume infusion should be considered for venoarterial ECMO, if available.

A FINAL WORD

The Adaptable Human

One of the notable features of heat-related illness and accidental hypothermia is the relatively low number of fatalities. For example, there were an average of 702 deaths yearly in the United States from heat-related illness over the years from 2004 to 2018 (31), compared to 695,000 deaths yearly from heart disease (32), and a 10-year survey in New York City (from 2009 to 2019) showed only 139 deaths from hypothermia, with an annualized death rate of 1.7 per million (33). These relatively small numbers are a testament to the ability of humans to adapt (both physiologically and behaviorally) to environmental extremes.

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A L G r a w a n y

Chapter 44

Fever in the ICU

Give me the power to produce fever, and I will cure all diseases.

Parmenides (ca. 500 B.C.)

The appearance of a new fever is always a matter of concern in a hospitalized patient. This chapter presents a practical approach to a new-onset fever in the ICU, and includes the preferred methods of measuring body temperature and obtaining blood cultures, and the most likely sources of fever in ICU patients, both noninfectious and infectious (1). The final section of the chapter focuses on the practice of suppressing fever, and why Parmenides would disapprove of this practice.

FEVER BASICS

Two scales (Celsius and Fahrenheit) are used to record body temperature, and the conversion from one scale to the other is shown in Table 44.1. Readings on the Celsius scale are often called "degrees centigrade", but this unit is intended for the degrees on a compass, not for temperatures (2).

Thermometry

. The preferred method of measuring body temperature is a thermistor-equipped catheter placed in either the pulmonary artery, esophagus, or urinary bladder (1). These catheters provide the most accurate measure of core body temperature, and they also allow continuous temperature monitoring.

. If the above methods are not available, then oral or rectal temperatures are preferred (1). However, oral temperatures can be 0.5° C (0.9° F) lower than core body temperatures in patients with tachypnea (3).

. Peripheral temperature measurements (i.e., tympanic membrane, temporal artery, and axillary temperatures) do not have clinically acceptable accuracy (4), and should not be used to make decisions based on the body temperature.

Definition of Fever

The body temperature has a diurnal variation, with the nadir in the early morning (between 4 and 8 AM) and the peak in the late afternoon (between 4 and 6 PM). The range of diurnal variation varies for each patient, and can be as high as 1.3° C (2.4° F) (5). Fever is best defined as a temperature that exceeds the normal diurnal variation for each patient, but this is not useful clinically (since the diurnal variation for each patients is usually not known). The following definitions of fever are included in the most recent clinical practice guidelines on fever in the ICU (1):

. In ICU patients, a single temperature measurement of 38.3° C (101° F) or higher represents a fever, and deserves further evaluation.

. A lower threshold of 38.0° C (100.4° F) can be used for neutropenic patients if the temperature elevation is sustained for at least one hour.

TABLE 44.1 Converting Celsius and Fahrenheit Temperatures

°C	°F	Conversion Formulas
100	212	Conversions are based on the corresponding temperatures at the freezing point of water,
40	104	0° C = 32° F
39	102.2	and the corresponding temperature ranges from freezing point to boiling point of water,
38	100.4	Δ100° C = Δ180° F
37	98.6	which can be reduced to:
36	96.8	Δ5° C = Δ9° F
35	95	These relationships are then combined to derive the following formulas:
34	93.2	0° F = (9/5° C) + 32°
33	91.4	0° C = 5/9 (° F – 32°)
0	32	

The Febrile Response

Fever is the result of inflammatory cytokines (called endogenous pyrogens) that act on the hypothalamus to elevate the body temperature. Therefore, any condition that triggers a systemic inflammatory response will produce a fever. Some implications of the febrile response are stated below.

. *Fever is a sign of inflammation, not infection*, and about 50% of ICU patients who develop a fever have no evidence of infection (6,7).

. The presence or severity of fever does not correlate with the presence or severity of infection. High fevers can be the result of a noninfectious process, while fever can be absent in life-threatening infections.

The distinction between inflammation and infection is an important one, not only for the evaluation of fever, but also for curtailing the unnecessary use of antibiotics.

Fever as an Adaptive Response

Whereas hyperthermia is the result of abnormal temperature regulation (see Chapter 43), fever is

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characterized by a normal thermoregulatory system that is operating at a higher set point (8). The consensus view is that fever is not a pathological condition, but rather is an adaptive response that enhances inherent defenses against infection (9). The beneficial effects of fever are described later in the chapter.

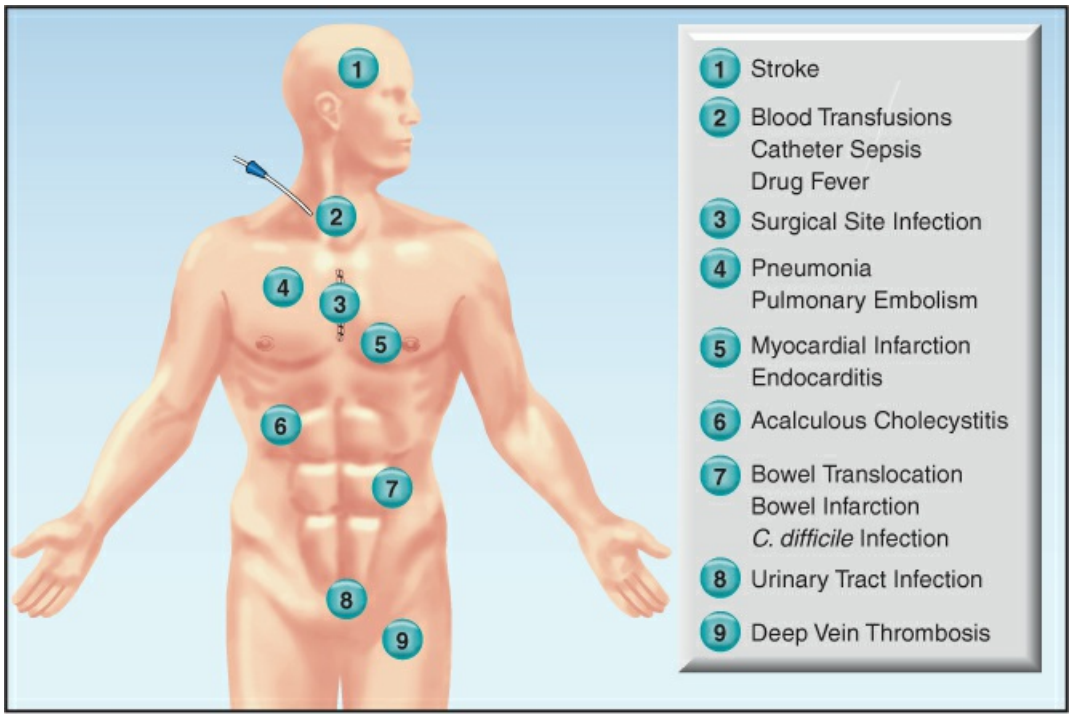


FIGURE 44.1 Potential sources of new-onset fever in the ICU.

Sources of Fever

Any condition capable of triggering an inflammatory response is capable of causing a fever. The notable sources of nosocomial fever in the ICU are shown in Figure 44.1.

NONINFECTIOUS SOURCES OF FEVER

As mentioned earlier, infection is responsible for only half of ICU-acquired fevers (6.7). The noninfectious conditions listed in Table 44.2 are responsible for most of the remaining fevers.

TABLE 44.2 Noninfectious Causes of Fever in the ICU

More Common	Less Common
Early Postoperative Fever	Adrenal Insufficiency
Venous Thromboembolism	Bowel Infarction
Drug Fever	Dressler Syndrome
Blood Transfusions	Fat Embolism
Myocardial Infarction	Tumor Lysis Syndrome
Alcohol/Drug Withdrawal	Malignant Hyperthermia
Inflammatory Lung Injury	Serotonin Syndrome

Ischemic Stroke	Iatrogenic Fever

Early Postoperative Fever

Major surgery is a source of tissue injury. Because inflammation and fever are the normal response to tissue injury, it is no surprise that fever in the first postoperative day is reported in 15–40% of cases of major surgery (10–12). In most cases, there is no evidence of infection (10,11), and the fever resolves in 24 to 48 hours.

Atelectasis Does NOT Cause Fever

There is a longstanding misconception that atelectasis is responsible for early postoperative fevers. One possible source of this misconception is the high incidence of atelectasis in patients who develop a postoperative fever. This is demonstrated in Figure 44.2 (see the graph on the left), which is from a study involving patients who underwent open heart surgery (12). Close to 90% of the patients with a fever on the first postoperative day had radiographic evidence of atelectasis. This, however, is *not* evidence that the atelectasis is the source of fever. In fact, the graph on the right in Figure 44.2 shows that most (75%) of the patients with atelectasis did not have a fever. The inability of atelectasis to produce fever was demonstrated over 50 years ago in an animal study where lobar atelectasis produced by ligation of a mainstem bronchus was not accompanied by fever (13).

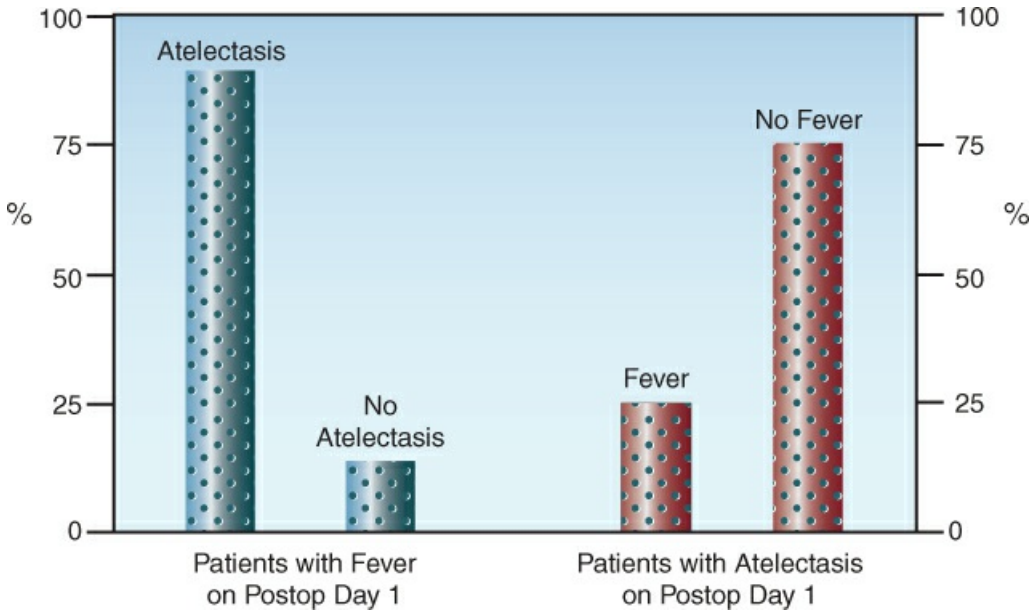


FIGURE 44.2 The relationships between fever and atelectasis in the first postoperative day following open heart surgery for 100 patients. The graph on the left shows that most patients with fever had atelectasis, but the graph on the right shows that most patients with atelectasis did not have a fever. Data from Reference 12.

To summarize, atelectasis is a common complication of major surgery, and occurs in over 90% of cases of general anesthesia (14). However, it is not a common cause of postoperative fever. *Most fevers that appear in the first 24 hours after surgery are the result of the tissue injury sustained during the procedure.*

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Malignant Hyperthermia

An uncommon but treatable cause of elevated body temperatures in the immediate postoperative period is *malignant hyperthermia* (15), an inherited disorder characterized by hyperpyrexia (temperature >104° F), muscle rigidity, and rhabdomyolysis in response to halogenated inhalational anesthetics. This disorder is described in Chapter 43.

Venous Thromboembolism

There are numerous risk factors for venous thromboembolism in ICU patients (see Table 5.1 and Figure 5.1), but the risk is highest in trauma victims and postoperative patients. Hospital-acquired deep vein thrombosis is usually asymptomatic, but acute pulmonary embolism can produce a fever that lasts for one week (16). The diagnostic approach to acute pulmonary embolism is described in Chapter 22.

Blood Transfusions

Transfusions of red blood cells (RBCs) and platelet products can trigger febrile reactions.

RBC Transfusions

Nonhemolytic fever is the most common adverse reaction to RBC transfusions (see Table 12.5), and usually appears during the transfusion, or within 6 hours after the transfusion is completed. These reactions are the result of antileukocyte antibodies in the recipient that react with donor leukocytes, and they are more likely to occur in patients who have received multiple transfusions.

Platelet Transfusions

Nonhemolytic fever is one of the more cited adverse reactions to platelet transfusions, especially with single-donor platelets (see Table 13.4). However, these reactions are infrequent (e.g., about one per 1,000 units transfused for single-donor platelets) (17).

Drug Fever

Drug-induced fever is typically a hypersensitivity reaction, but can also be an idiosyncratic reaction. Any drug can potentially trigger a febrile response, but the ones reported most frequently are listed in Table 44.3 (18,19).

There is no standard description for drug fever. The onset of the fever varies from a few days to more than a month after the start of drug therapy, and the fever can appear as an isolated finding, or it can be accompanied by the other manifestations listed in Table 44.3 (20). Note that about half of the patients have rigors, and about 20% develop hypotension, indicating that *patients with a drug fever can appear to be seriously ill*. Evidence of a hypersensitivity reaction (i.e., eosinophilia and rash) is absent in over 75% of cases of drug fever (20).

TABLE 44.3 Drug-Associated Fever

Common Offenders[a]	Clinical Findings[b]
Amphotericin	Rigors (53%)

Carbamazepine	Myalgias (25%)
Cephalosporins	Leukocytosis (22%)
Heparins	Eosinophilia (22%)
Penicillins	Rash (18%)
Phenytoin	Hypotension (18%)
Procainamide	
Quinidine	

[a]For a more complete list, see Reference 18. [b]From Reference 20.

Suspicion of drug fever usually occurs when there are no other likely sources of fever. When suspected, possible offending drugs should be discontinued, and the fever should disappear within 72 hours (18).

Other Sources

The following are other some noninfectious causes of fever that deserve mention:

. Severe withdrawal from alcohol, opiates, benzodiazepines, barbiturates, and possibly gabapentin, can cause a fever that typically appears a few days after abstinence.

. Adrenal insufficiency is a recognized cause of fever, and can be the result of spontaneous adrenal hemorrhage from anticoagulant therapy or disseminated intravascular coagulation (DIC). The diagnosis of adrenal insufficiency is described in Chapter 51.

. Acute respiratory distress syndrome (ARDS) is an inflammatory lung injury from multiple sources that can produce a fever. This condition is described in detail in Chapter 24.

. Infarctions of the heart, brain, and bowel are potential sources of fever.

. The serotonin syndrome, which is accompanied by fever, can be a complication of treatment with critical care drugs like fentanyl, tramadol, and linezolid (see Table 43.4).

Iatrogenic Fever

Faulty thermal regulators in water mattresses and aerosol humidifiers can cause fever by transference (21). It takes only a minute to check the temperature settings on heated mattresses and ventilators, but it can take far longer to explain why such a simple cause of fever was overlooked.

NOSOCOMIAL INFECTIONS

The major ICU-acquired infections are pneumonias (especially ventilator-associated pneumonias), bloodstream infections from vascular catheters, urinary tract infections, surgical site infections, and *Clostridium difficile* infections: Table 44.4 shows the number of each of these infections reported in the United States over a one-year period (22). All but one of these infections is described elsewhere in the book: i.e., catheter-related bloodstream infections are described in Chapter 3, ventilator-associated pneumonia in Chapter 29, *C. difficile* infections in Chapter 41, and urinary tract infections in Chapter 42.

Surgical Site Infections

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Surgical site infections (SSIs) continue to be a significant source of postoperative morbidity despite a long history of preventive efforts. These infections typically appear 5–7 days after surgery, but any wound infection that develops within 30 days following surgery is classified as an SSI (23). Infected wounds are identified by signs of local inflammation (erythema, edema, tenderness), and by purulent or foul-smelling drainage. Superficial infections are less likely to produce fever than infections with deep tissue involvement.

There are multiple risk factors for SSIs, including the condition of the patient and the type of procedure. The reported incidence of SSIs is 1–2% following clean procedures (where the chest or abdomen is not opened), and 4–5% following contaminated procedures (e.g., bowel perforation with soilage of the peritoneal cavity) (24). The pathogens involved in SSIs are also determined by the surgical procedure: e.g., SSIs from clean surgical procedures usually involve *Staphylococcus aureus*, while SSIs from contaminated procedures often involve gram-negative aerobic bacilli (24).

TABLE 44.4 Hospital-Acquired Infections in the United States in 2021

Infection	Total #	# in ICUs
Ventilator-Associated Events	50,050	47,254
C. difficile Infection	44,948	—
Catheter-Associated Bloodstream Infection	27,021	14,003
Catheter-Associated Urinary Tract Infection	24,710	12,208
Surgical Site Infection	21,186	—

*A ventilator-associated event is a respiratory infection (tracheobroncitis or pneumonia) that appears after 48 hours of mechanical ventilation, and results in an increased requirement for inhaled O_2 or PEEP. Data from Reference 21.

Management

Management of SSIs includes debridement, with antibiotics added if there is a fever. For initial antibiotic coverage, vancomycin is appropriate if *Staphylococcus aureus* is suspected, and for contaminated procedures (e.g., abdominal surgery), either *piperacillin/tazobactam*, or a third generation cephalosporin (e.g., *ceftriaxone*) plus *metronidazole* is advised (25).

Necrotizing Wound Infections

Necrotizing wound infections are evident in the first few postoperative days. Early signs include marked erythema and edema, and *pain that is out of proportion to the skin changes* (due to ischemic nerve injury) (26). Crepitance and skin blistering are later findings. Spread to deeper structures (with rhabdomyolysis) is rapid, and a milky or foul-smelling discharge may be evident (from the underlying necrosis). These infections can be caused by a single organism (usually β-hemolytic streptococci or *Clostridium* species), or they can be polymicrobial (usually a mix of gram-negative enteric organisms and anaerobes) (26).

MANAGEMENT: *Emergent and extensive debridement is mandatory for survival*, along with appropriate antibiotic therapy. A Gram stain can help with the initial antibiotic selection. If

streptococci or clostridial organisms are suspected, the appropriate regimen is *penicillin plus clindamycin*, and if the surgery involved the deep structures in the abdomen or thorax, then a carbapenem (e.g., *meropenem*) is advised (25).

Other Considerations

The following are infections that should be considered in specific situations.

Endocarditis

Nosocomial endocarditis is uncommon, but an evaluation for endocarditis is indicated in the following situations:

. Any patient with bacteremia involving streptococci, staphylococci, or enterococci, who also has an abnormal or prosthetic heart valve.

. All patients with blood cultures growing *Staphylococcus aureus*, regardless of the condition of the heart valves.

DIAGNOSIS: Echocardiography is the standard method used to diagnose endocarditis. The evaluation can begin with *transthoracic echocardiography* (TTE), which is easily obtained at the bedside. However, TTE has a limited sensitivity for the detection of vegetations on native valves (sensitivity of 60%–70%), and especially on prosthetic valves (sensitivity of 50%) (27), so a negative TTE must be followed by *transesophageal echocardiography* (TEE), which has a sensitivity of at least 90% for the detection of vegetations on native and prosthetic valves (27).

THERAPY: The antimicrobial therapy for endocarditis has several variations (depending on the infecting organism, patterns of resistance, and the involvement of a native or prosthetic valve), and it is a prolonged affair that can be marred by complications (e.g., emboli) (28). It thus seems wise to have an infectious disease specialist guide the treatment of these infections.

Peritoneal Infections

. Peritonitis or intra-abdominal abscess should be suspected in any patient who has had extensive or contaminated abdominal surgery and has a persistent or late-onset (one week or later) postoperative fever. These infections are described in Chapter 41.

. Spontaneous bacterial peritonitis is a consideration in any patient with cirrhosis and ascites who develops an unexplained fever. This infection is described in Chapter 39.

Disseminated Candidiasis

Risk factors for disseminated candidiasis include indwelling central venous catheters, recent abdominal surgery, or recent exposure to broad spectrum antibiotics. Most ICU patients have at least one of these risk factors, so *disseminated candidiasis should be considered in the following situations*:

. Any ICU patient with persistent fever of unclear etiology.

. A patient with a *Candida* urinary tract infection.

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DIAGNOSIS: Disseminated candidiasis is a challenging diagnosis because ICU patients are often colonized with *Candida* species, and the sensitivity of blood cultures for the detection of invasive *Candida* infections is as low as 20% (29). However, *a polymerase chain reaction (PCR)-based test is commercially available that can detect Candida species in blood with a sensitivity and specificity >90%* (30), and this test should be considered (if available) for patients with possible invasive candidiasis. Persistent candiduria or a *Candida* urinary tract infection is usually a consequence of disseminated candidiasis (31), so this can be a valuable sign of invasive disease.

ANTIFUNGAL TREATMENT: About half of the cases of disseminated candidiasis are due to *Candida* species other than *Candida albicans*, so fluconazole is not appropriate for empiric coverage. Instead, empiric antifungal coverage should include an echinocandin (i.e., *micafungin, caspofungin,* or *anidulafungin*) or *amphotericin* for severely ill patients.

GENERAL CONSIDERATIONS

The evaluation of fever can travel many roads, but the following measures are included in most evaluations.

Blood Cultures

Blood cultures are recommended for all cases of ICU-related fever where an infection is possible. The yield from blood cultures is dependent on the volume of blood withdrawn during a venipuncture, and the number of venipuncture sites.

Influence of Volume

The yield from blood cultures is optimal when 20–30 mL of blood is withdrawn from each venipuncture site (32). The standard practice is to withdraw 20 mL of blood from a venipuncture site: one-half (10 mL) is then injected into each of the two bottles of broth (one aerobic and one anaerobic) provided in a blood culture set. Increasing from 20 ml to 30 mL of blood increases the yield from blood cultures by about 10% (33). When using 30 mL per venipuncture, the extra 10 mL aliquot of blood should be injected into the aerobic broth bottle (33).

Number of Blood Cultures

The relationship between the number of blood cultures and the detection of bacteremia is shown in Figure 44.3 (34). (*Note:* The number of blood cultures represents the number of venipuncture sites.) This is from a study of patients with bacteremia documented by four or more blood cultures drawn over a 24 hour period. Note that the majority of the bacteremias (94%) were detected with two blood cultures in the patients with endocarditis, while three blood cultures were required to detect over 90% of the bacteremias in patients with other infections. This data is the basis for the following recommendations:

. Three to four blood cultures drawn over a 24-hour period is recommended for the optimal detection of bacteremia (1).

. Two blood cultures are satisfactory for detecting bacteremia in cases of endocarditis.

The bacteremia in endocarditis is continuous, which may explain why fewer blood cultures are needed to detect bacteremia.

Multilumen Catheters

The following recommendation pertains to the evaluation of possible catheter-related septicemia with multilumen central venous catheters (1).

. For multilumen catheters, separate blood cultures should be drawn through two lumens of the catheter, along with a blood culture from a peripheral venipuncture site. *Note:* Cultures from a single port can miss as many as 40% of catheter-related infections (35).

. If the same organism is isolated in catheter and peripheral blood, then the "time to positivity" is compared. When growth in catheter blood appears at least two hours before growth in peripheral blood, the diagnosis is catheter-related septicemia.

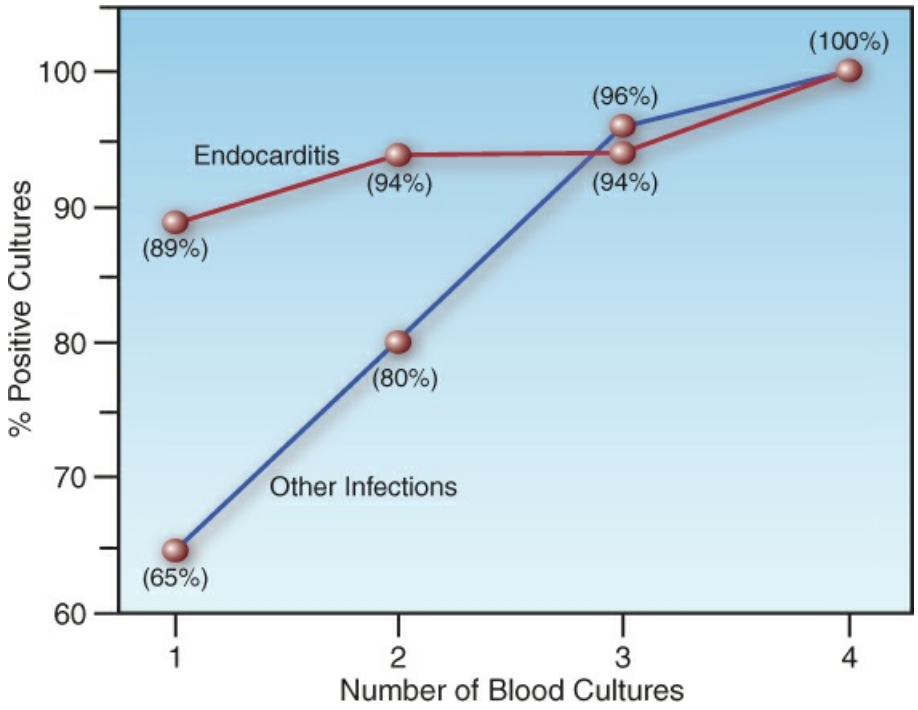


FIGURE 44.3 Relationship between the number of blood cultures drawn over a 24 hour period (20 mL per blood culture) and the detection rate for bacteremia. See text for explanation. Data from Reference 34.

Each lumen of a multilumen catheter is a potential source of septicemia (36). so the optimal strategy would be to culture all ports of the catheter. (The diagnosis of catheter-related septicemia is described in detail in Chapter 3.)

Molecular Testing

Growth in blood cultures often takes 48 hours or longer to detect, which has disadvantages (e.g., it delays treatment with appropriate antibiotics when cultures are positive, and promotes the overuse of antibiotics when cultures are negative). More rapid detection of organisms in the bloodstream is possible using DNA amplification (like the PCR method mentioned earlier for

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detecting candidemia), and a commercially available product is available (T2Bacteria® Panel, T2Biosystems, Lexington MA) that can detect six bacterial species (*E. coli, P. aeruginosa, K. pneumoniae, S. aureus, E. faecium,* and *A. baumannni*) with a sensitivity and specificity of 90% (37). Results can be available in 3–5 hours.

The value of the rapid detection methodology for improving outcomes has yet to be determined, but seems questionable, since a positive test still requires culture results for antibiotic sensitivities, and a negative test does not rule out infection with other organisms.

Empiric Antimicrobial Therapy

Empiric antibiotic therapy is recommended when the likelihood of infection is high. Prompt initiation of antimicrobial therapy is considered essential, *particularly in patients with neutropenia* (absolute neutrophil count <500), where delays of only a few hours can have a negative impact on outcomes (38).

Empiric antibiotic coverage is described elsewhere in the book for ventilator-associated pneumonia (see Table 29.6), catheter-related bloodstream infections (see Table 3.6), catheter-associated urinary tract infections (see Chapter 42), and surgical site infections (see earlier in this chapter). The following are some general statements about empiric antibiotics for ICU-related fever.

. Most ICU-related infections are caused by *Staphylococcus aureus*, or bowel microbes (e.g., *E. coli, P. aeruginosa, C. difficile, Enterococcus* spp).

. Empiric coverage for the major ICU-related infections (i.e., pneumonia, vascular catheter infections, urinary tract infections, postop abdominal infections) should include an antibiotic that is active against gram-negative aerobic bacilli. Popular choices are *piperacillin/tazobactam, cefepime,* or *meropenem*.

. Coverage for *S. aureus* should be included for suspected catheter-related infections and ventilator-associated pneumonia. Vancomycin is a popular choice because it covers both methicillin-resistant and methicillin-sensitive strains of *S. aureus*.

. An antifungal agent should be considered for patients who have a risk of invasive candidiasis (e.g., those with central venous catheters) who also have: (*a*) persistent fever of unclear etiology, (*b*) fever following contaminated bowel surgery, or (*c*) fever associated with persistent candiduria. An agent that covers all *Candida* species (such as an echinocandin) is most appropriate.

ANTIPYRETIC THERAPY

The popular perception of fever as a malady that must be corrected is rooted in hearsay. In fact, it seems that *fever is a normal adaptive response that enhances the ability to eradicate infection* (9). This section contains some observations about fever that are intended to make you think twice about starting antipyretic therapy in critically ill patients.

Fever as a Host Defense Mechanism

An increase in body temperature can enhance immune function by increasing the production of

antibodies and cytokines, activating T-lymphocytes, facilitating neutrophil chemotaxis, and enhancing phagocytosis by neutrophils and macrophages (39,40). Fever also inhibits bacterial growth and viral replication. This is demonstrated in Figure 44.4 (41), which shows that an increase in temperature of 4° C completely suppresses growth of the microorganism.

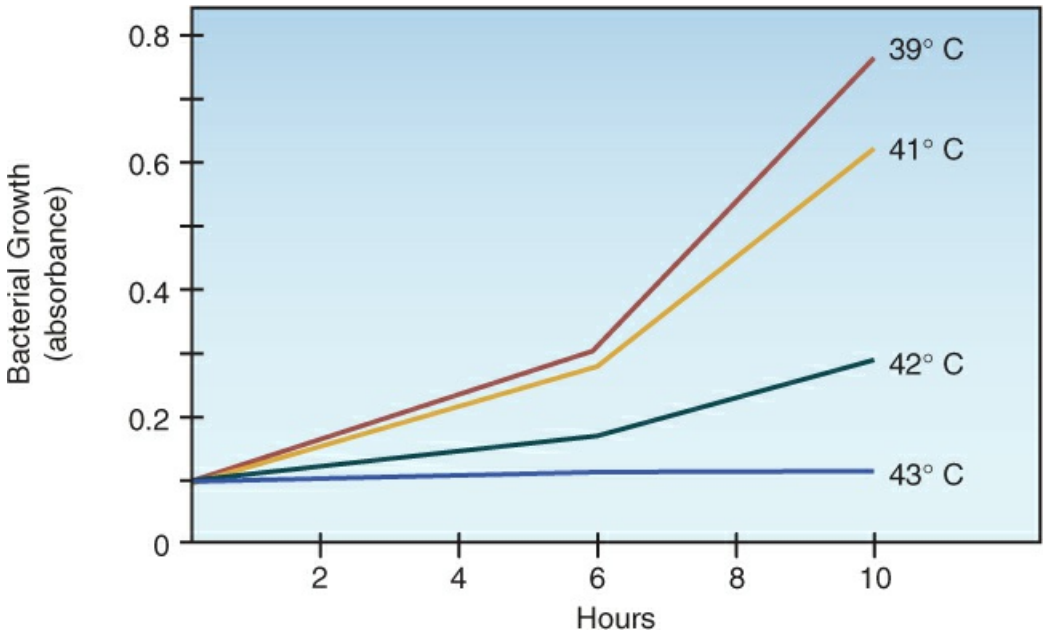


FIGURE 44.4 The influence of body temperature on the growth of *Pasteurella multocida* in the blood of infected laboratory animals. The range of temperatures in the figure is the usual range of febrile temperatures for the study animal (rabbits). Data from Reference 41.

Clinical Studies

The benefits of fever as a host defense against infection is supported by clinical studies showing an inverse relationship between body temperature and survival in patients with severe sepsis and septic shock (42–44). The results of one of these studies is shown in Figure 44.5. In this case, the initial body temperature showed an inverse relationship with in-hospital mortality in patients admitted to the ICU with severe sepsis or septic shock (42).

Similar results have been reported in other studies (43,44), including a study that showed a higher mortality rate in septic patients who received antipyretic therapy (45).

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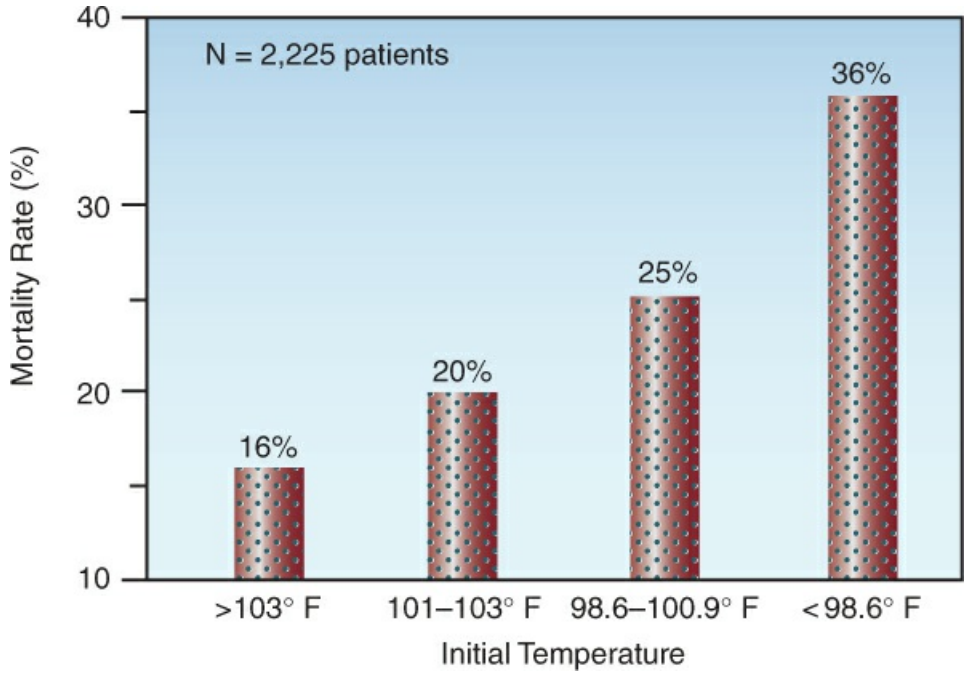


FIGURE 44.5 Inverse relationship between body temperature and in-hospital mortality rate in 2,225 patients with severe sepsis and septic shock. Differences between categories are significant at the p<0.001 level. Data from Reference 42.

THERAPEUTIC HYPERTHERMIA: The evidence of improved outcomes with fever has prompted a study in which afebrile patients with sepsis were treated with external warming to raise their body temperature by 1.5° C, and this resulted in a significant decrease in the mortality rate when compared to afebrile patients who were not warmed (46).

Summation

The available evidence shows that fever is not a pathological condition, but rather is normal adaptive response that serves an antimicrobial defense mechanism. The changing perception of fever is evident in the most recent clinical practice guidelines on fever in the ICU, which recommends not suppressing fever as a routine practice (1).

Note: It is important to mention that fever *can* be harmful (and thus should be suppressed) in patients with ischemic brain injury from a cardiac arrest or acute stroke.

Antipyretic Drugs

Prostaglandin E mediates the febrile response to endogenous pyrogens, and drugs that interfere with prostaglandin E synthesis are used as antipyretic agents. These drugs include aspirin, acetaminophen, and nonsteroidal anti-inflammatory agents (NSAIDs), but only the latter two are used for antipyresis in the ICU.

Acetaminophen

Acetaminophen is an effective antipyretic agent in critically ill patients (47), and is the most popular drug used for fever suppression in ICUs.

DOSING REGIMENS: Acetaminophen can be given orally or by rectal suppository in a dose of 650 mg every 6 hours, and an intravenous preparation is available that can be given in a dose of 650 mg every 6 hours, or 1,000 mg every 8 hours. *Caution:* Acetaminophen is hepatotoxic, and *the maximum daily dose that is considered safe is 3 grams* (FDA announcement, January 13, 2011). The drug is contraindicated in patients with hepatic insufficiency. (Acetaminophen hepatotoxicity is described in Chapter 52.)

NSAIDs

Two NSAIDs have been evaluated for fever suppression in critically ill patients: *ibuprofen* and *ketorolac*. Ibuprofen is effective in an intravenous dose of 10 mg/kg, up to 800 mg every 6 hours (48), and ketorolac is effective in a single intravenous dose of 0.5 mg/kg (49). There is a limited experience with NSAIDs in the ICU, largely due to an exaggerated fear of renal toxicity and GI hemorrhage from the use of these drugs.

Cooling Blankets

External cooling is a popular antipyretic measure, but is counterproductive in the setting of fever. The febrile response raises the body temperature by promoting cutaneous vasoconstriction and increasing skeletal muscle activity (via rigors and shivering). This is what the body normally does in response to a cold environment, so the febrile response mimics the physiological response to cold. Stated another way, *the febrile response makes the body behave like it is wrapped in a cooling blanket*. Adding a cooling blanket will only aggravate the cutaneous vasoconstriction and increased muscle activity (via shivering) that produces the fever. Cooling blankets are more appropriate for heat stroke, where the thermoregulatory response to a warm environment is lost.

A FINAL WORD

The 1927 Nobel Prize

The 1927 Nobel Prize in Medicine was awarded to Julius Wagner-Jauregg (an Austrian physician) for his discovery of the therapeutic benefits of fever in patients with neurosyphilis. Warner-Jauregg had observed that remissions in dementia paralytica (neurosyphilis) were often preceded by a febrile illness, and he subsequently performed a bold experiment where he inoculated 9 neurosyphilis patients with tertian malaria (to induce a recurrent fever). Surprisingly, 6 of the 9 patients experienced extensive remissions, with almost complete resolution of symptoms in 3 patients (50). These results were corroborated in other patients, along with the discovery that spirochetes disappeared from brain tissue during the periods of fever, and reappeared when the fever subsided.

The practice of inoculating patients with malaria to treat neurosyphilis was eventually abandoned, because it introduced a disease to treat a disease. Unfortunately, the implications of Wagner-Jauregg's work (i.e., that fever can eradicate infection) were also abandoned.

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General Considerations

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A Final Word

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